# Locally spherical regular hypertopes: marked quotients, affine families and a corrected finite table

Human Compression project

**Abstract**

We give a complete attributed account of the fixed revised manuscript *An exploration of locally spherical regular hypertopes* by Fernandes, Leemans and Ivić Weiss. The spherical classification follows from Maxwell's retained normal-subgroup theorem and explicit antipodal incidence arguments. A reflection-commutator proof gives the affine translation-kernel conclusion under the universal-residue hypothesis, and the displayed toroidal families are joined to their exact parameter domains. A finite separating-set argument supplies arbitrarily large finite examples for every retained compact hyperbolic diagram of rank at least three. Complete bounded quotient certificates verify twenty-two named rows of the source's finite table and exclude its other four rows. We state both diagram-only and universal-residue minimum conventions. Auxiliary quotient assertions are corrected explicitly; classifications and earlier constructions retained from the literature are identified throughout.

## 1 Scope, conventions and acquired proofs

Our source is the sixteen-page revised manuscript [1], whose fixed PDF identity and complete clause ledger accompany this article. Its page 5 defines a spherical hypertope by a finite Coxeter diagram; this allows projective quotients. Section 5, page 10, expressly restricts instead to universal spherical residues. We call the first condition *broad* and the second *locally universal*. Neither convention is silently imposed on the source's hyperbolic table. Isomorphism in the source permits reordering types, including duality. Counts of normal kernels for an ordered presentation are a different invariant and will never be called counts of unlabelled hypertopes.

A hypertope is a thin, residually connected incidence geometry. Its group $\mathrm{Aut}_I$ of type-preserving automorphisms is distinguished from its group allowing type permutations. Regularity means that $\mathrm{Aut}_I$ acts regularly on chambers. For an involutory marking $G = \langle r_i : i \in I \rangle$ write $G_J = \langle r_j : j \in J \rangle$ and $P_i = G_{I\setminus\{i\}}$. The intersection property is $G_J \cap G_K = G_{J\cap K}$ for all $J, K \subseteq I$.

The complete arguments acquired in this article are supplied as separate editable manuscripts in the reading volume, with their finite certificates and distinct reviews. They are: the residue and marking theorem [15]; the quotient-kernel correction [16]; the spherical argument [17]; the affine kernel and period arguments [18, 19]; finite residue-preserving quotients [20]; the eight-diagram finite core and named groups [21, 22]; the exact Coxeter/LINS interfaces [23]; the ninth-square and exact-order exclusions [24, 25]; and the infinite cubic example [26]. The present proofs make the remaining elementary source joins. Acquiring these complete proofs does not reprove their explicitly stated classical or computational premises.

## 2 Markings, cosets and the quotient correction

We use the proved residue theorem [15]: the distinguished chamber involutions of a regular hypertope generate its full type-preserving group, satisfy the intersection property, and identify all flag stabilizers with the corresponding marked subgroups. Conversely a C-group whose actual coset incidence system is flag transitive gives a regular hypertope. This is the interface of source

Theorem 2.3; the string C-group/polytope correspondence in source Theorem 2.4 remains the classical Aschbacher–Schulte result.

**Proposition 1.** *For any family $(P_i)_{i\in I}$ of subgroups of a group $G$, the typed right cosets $P_ig$, with incidence given by nonempty intersection, form an incidence system with base chamber $(P_i)_{i\in I}$. Right multiplication acts transitively on flags of each prescribed type of size at most two. Its kernel is $\bigcap_i \mathrm{Core}_G(P_i)$.*

*Proof.* Cosets of the same subgroup are equal or disjoint, which verifies the same-type incidence axiom. All base cosets contain the identity. Multiplication preserves intersections. For an incident pair choose an element $g$ in its intersection; the pair is then $(P_ig, P_jg)$. This also proves transitivity for singleton flags. An element fixes every coset of type $i$ precisely when it lies in every conjugate of $P_i$. Intersecting this condition over the types gives the asserted kernel. □

Thus the sentence preceding source Proposition 2.2 must not identify an arbitrary $G$ with the full automorphism group. The proposition's actual action statement is valid. For a regular hypertope the stabilizer/coset theorem supplies the faithful full group, not this general construction alone.

Source Proposition 2.1 and Theorem 3.1 are proved in [16]. In particular, C-group maximal parabolics and all the expected pairwise maximal intersections imply the full intersection property. An epimorphism to a C-group that is injective on all maximal marked parabolics therefore lifts that property. This implication points *upward*; it does not give a reverse homomorphism from a quotient to its cover.

The exact downward criterion, under local injectivity, is

$$N \cap P_i = 1 \quad \text{for all } i, \qquad G/N \text{ is a C-group} \iff N \cap P_iP_j = \{1\} \quad (i \neq j). \tag{1}$$

Here $G$ itself is a C-group and $N \triangleleft G$; the proof lifts two equal coset images into $P_i$ and $P_j$ and applies the maximal-parabolic criterion. Flag transitivity must still be checked. The affine $\widetilde{A}_2$ example in [16] has $G = \mathbb{Z}^2 \rtimes S_3$ and $N = \mathbb{Z}^2$. Each maximal parabolic maps injectively onto $S_3$, yet the quotient incidence system has one object of each type and one chamber. It is flag transitive but not thin. This is a counterexample to source Theorem 3.2, and (1) is its precise intersection repair.

For any regular hypertope, its marked Coxeter cover $W \twoheadrightarrow G$ and coset reconstruction identify the hypertope with the incidence quotient of the universal complex. Local injectivity follows if its proper residues are universal, since their chamber groups have the same finite orders as the covering parabolics. It does not follow from a finite residue diagram alone. The order-1920 square in Section 6 has local groups of order 60, whereas the covering $H_3$ parabolics have order 120. It is an explicit broad counterexample to the local-injectivity clause of source Theorem 3.4. The marked quotient and incidence reconstruction survive.

**Proposition 2.** *A regular hypertope whose diagram has connected components $I_1, \dots, I_t$ is the direct sum of the corresponding regular hypertopes. Its distinguished group is $G_{I_1} \times \cdots \times G_{I_t}$.*

*Proof.* Marks in different components commute. The intersection property gives $G_{I_a} \cap G_{I\setminus I_a} = 1$. Multiplying the commuting factors therefore gives an injective homomorphism from their direct product onto $G$. For $i \in I_a$ the maximal parabolic equals the product of $(G_{I_a})_{I_a\setminus\{i\}}$ and all other factors. Its cosets are thus exactly the cosets in the $a$th factor. Cosets belonging to different components always intersect; within a component their incidence is its factor coset incidence. This proves the direct-sum description. Fixing the base objects outside $I_a$ identifies that factor with a residue of the original hypertope. It is consequently a regular hypertope by the acquired residue theorem. □

### The two geometric examples

For source Example 2.5, colour unit cubes of $\mathbb{Z}^3$ by the parity of their lower vertex. The exact dual-cell construction [26] sends the two cube colours to even and odd vertices of the shifted dual grid, source edges to dual squares, and source vertices to dual cubes. Source types $(0, 2, 1, 3)$ become types $(0, 1, 2, 3)$. The direct infinite-grid proof gives flag extension, thinness, residual connectivity and a regular chamber action of $D \rtimes B_3$, where $D = \{v \in \mathbb{Z}^3 : \sum v_i \equiv 0 \pmod 2\}$. Its marks are $\tau_{(1,1,0)} z_1 z_2 s_1, s_1, s_2, z_3$, with $z_i$ sign changes and $s_i$ adjacent coordinate swaps. This is the zero-period object itself; it is not obtained by setting a finite parameter restricted to $t \geq 2$ to zero.

For precision, the negative flag in [17] also proves the nongeometry assertion of source Example 3.3. In the usual $D_4$ model let $v = (1, 1, 1, 1)/2$, $w = (1, 1, 1, -1)/2$, and $x = e_4$. Their antipodal classes are pairwise incident. A fourth-type object is represented by $y = \pm e_i \pm e_j$. Incidence with $[x]$ requires that one coordinate be 4. Incidence with $[v]$ then requires the two signs of $y$ to agree, whereas incidence with $[w]$ requires them to disagree. Thus this flag has no fourth-type extension. This strengthens the same signed-triangle obstruction from failure of the proposed regular action to failure of flag extension in this rank-four example.

## 3 The complete spherical classification

Retain Maxwell's Theorem 0.1 [2], as quoted in source Theorem 4.1: a normal subgroup of an irreducible finite Coxeter group is the central antipodal subgroup or the kernel of a graph homomorphism. The primary full proof is not acquired. A graph homomorphism sends simple generators to simple generators or to the identity and covers the target simple generators. In a hypertope quotient the marks remain distinct, nontrivial, and preserve every pair order. Such a graph homomorphism is an isomorphism of the defining Coxeter presentations. The only remaining kernel is the central antipode, when present.

The full argument [17] proves that the antipode avoids every proper parabolic and every product $P_i P_j$. It therefore gives a C-group quotient. The string correspondence pays its geometry for string diagrams. For the other cases, fundamental-weight inner products determine incidence; every quotient flag lifts exactly when its signed triangles have positive sign product. A single explicit triangle excludes every even $D_n$ case. For $E_7, E_8$, fifteen complete, generator-closed weight orbits, containing 899402 vectors, exclude every bad triangle. Their accepted certificates are acquired, not enumerated again here. The resulting classification is:

| Diagram | Range | Universal group order | Additional antipodal object |
|---|---|---|---|
| $A_n$ | $n \geq 1$ | $(n+1)!$ | no |
| $D_n$ | $n \geq 4$ | $2^{n-1} n!$ | no |
| $B_n$ | $n \geq 3$ | $2^n n!$ | yes |
| $I_2(p)$ | $p \geq 3$ | $2p$ | no |
| $E_6$ | | 51840 | no |
| $E_7$ | | 2903040 | yes |
| $E_8$ | | 696729600 | yes |
| $F_4$ | | 1152 | yes |
| $H_3$ | | 120 | yes |
| $H_4$ | | 14400 | yes |

An additional antipodal object has half the displayed group order. All proper residues of every object in this table are universal spherical, by the embedded parabolics and exact residue reconstruction. This proves source Theorems 4.2–4.3, subject to the named retained inputs, without assuming local injectivity in order to obtain it.

The $E_6$ entry corrects source Table 1's $12 \cdot 6!$ to $72 \cdot 6!$. Its exact certificate has a 27-element fundamental-weight orbit with standard $D_5$ stabilizer of order 1920, giving $27 \cdot 1920 = 51840$.

Conventional extended Schläfli notation in source Table 1 is retained as notation for these universal/projective objects. No additional presentation theorem is inferred from that notation. The source's blanket half-power description of a longest element is not used: in standard $A_3$, $c = (1234)$ has $c^2 = (13)(24)$, while the longest element is $(14)(23)$. Only actual central antipodes, with the exact powers verified in [17], occur in the proof.

## 4 Affine kernels and the displayed families

Let $W$ be an irreducible affine Coxeter group in its faithful Euclidean action, and let $N \triangleleft W$ meet every maximal standard parabolic trivially. The complete proof [18] first shows that $N$ is torsion-free: a finite-order isometry fixes a point, which can be moved into the closed alcove and hence has a proper parabolic stabilizer. For $g(x) = Ax + b \in N$, the commutator $gs_ig^{-1}s_i$ is the product of reflections in $gM_i$ and $M_i$. Nonparallel mirrors give a nontrivial finite-order element, so each wall-normal line is preserved by $A$. Connectedness of the nonorthogonality graph of the wall normals forces $A = I$ or $-I$. The latter makes $g$ an involution. Thus $N$ consists of translations. This also covers the one-dimensional case directly.

Consequently source *Corollary* 5.1 holds for the locally universal condition explicitly imposed in its Section 5: the hypertope is an incidence quotient by a normal translation subgroup. Maxwell's affine normal-subgroup classification is not required by this proof. The conclusion is necessary, not sufficient: arbitrary translation subgroups can fail (1) or full flag validity. The affine $A_2$ example above also qualifies the source's page 11 assertion that avoiding individual parabolics completes quotient classification. Euclidean diagrams with projective residues, such as the source's attributed Schulte/Hartley examples, remain outside this locally universal kernel conclusion.

The displayed family classifications are retained with their exact domains. Coxeter [5] and Cacciari [6] give, up to duality,

$$\{4,4\}_{(s,0)},\ \{3,6\}_{(s,0)}\ (s \geq 2), \qquad \{4,4\}_{(s,s)},\ \{3,6\}_{(s,s)}\ (s \geq 1),$$

and $(3,3,3)_{(s,0)}, (3,3,3)_{(s,s)}$ with $s \geq 2$ in both cases. McMullen–Schulte [7, 4] supply the cubic families $\{4, 3^{n-2}, 4\}_{(s^k,0^{n-k})}$ for $s \geq 2$, $k = 1, 2, n$, and the $F_4$ families $\{3,3,4,3\}_{(s^k,0^{4-k})}$ for $s \geq 2$, $k = 1, 2$. These are attributed classification inputs, not new lattice exhaustions.

For the two nonstring rank-four diagrams the literal adapter [19] proves existence and marked groups of all six displayed families. Put $E = 2\mathbb{Z}^3 + \mathbb{Z}(1,1,1)$; $B_3^+$ is the signed permutation subgroup with an even number of sign changes. Their groups are $(D/L) \rtimes B_3$ and $(D/L) \rtimes B_3^+$, respectively:

| Period and domain | $L$ | $\lvert G_{\widetilde{B}_3}\rvert$ | $\lvert G_{\widetilde{A}_3}\rvert$ |
|---|---|---|---|
| $(2s,0,0)$, $s \geq 2$ | $2s\mathbb{Z}^3$ | $192s^3$ | $96s^3$ |
| $(s,s,0)$, $s \geq 2$ | $sD$ | $48s^3$ | $24s^3$ |
| $(2s,2s,2s)$, $s \geq 1$ | $2sE$ | $768s^3$ | $384s^3$ |

Actual halving geometries, including the smallest allowed periods, supply thinness, residual connectivity and full regularity. The claim that these families exhaust the rank-four cases is retained from Ens [8]. The source's announced all-rank geometric classification in its reference 21 is an announcement, not a proved result acquired here.

## 5 Compact diagrams and finite existence

The finite, affine and compact hyperbolic simplex diagram classifications are retained from Humphreys [3] and McMullen–Schulte [4]. The exact twenty-one printed rows of source Table 2 are recorded in the clause ledger. Its ten affine rows are $\widetilde{A}_{n-1}$ $(n \geq 3)$, $\widetilde{D}_{n-1}$ $(n \geq 5)$, $\widetilde{C}_{n-1}$

$(n \geq 3)$, $\widetilde{B}_{n-1}$ $(n \geq 4)$, $\widetilde{E}_6, \widetilde{E}_7, \widetilde{E}_8, \widetilde{F}_4, \widetilde{G}_2, \widetilde{A}_1$. The last is rank two with an infinity bond. The eleven hyperbolic rows comprise the strings $[p, q]$ with $1/p + 1/q < 1/2$, and triangles $(k, l, m)$ with $1/k+1/l+1/m < 1$; the rank-four squares $(3, 3, 3, p)$, $p = 4, 5$, $(3, 4, 3, 4)$, $(3, 5, 3, p)$, $p = 4, 5$; strings $[p, 3, 5]$, $p = 4, 5$, and $[3, 5, 3]$; the fork $[5, 3_3]$; the rank-five pentagon $(3, 3, 3, 3, 4)$; strings $[p, 3, 3, 5]$, $p = 3, 4, 5$; and the fork $[5, 3, 3_3]$. All unspecified finite labels in this description are at least three. Repeated parameter values give nine rank-four and five rank-five diagrams.

**Theorem 3** (Acquired finite-existence theorem)**.** *If an infinite Coxeter system has finite rank at least three and every maximal standard parabolic is finite, it has finite regular hypertope quotients of unbounded chamber number with unchanged diagram and universal spherical proper residues.*

The proof [20] avoids every nonidentity element in the finite union $\bigcup_{i,j,k} P_i P_j P_k$. Two factors ensure the full intersection property; three factors ensure that pairwise incident quotient objects represented within one parabolic were incident before quotienting. Every flag then lifts and each full residue incidence is preserved. In the faithful geometric representation all coefficients lie in a fixed ring of algebraic integers. A rational prime avoiding finitely many nonzero norms separates this finite set by reduction. Separating arbitrarily many further elements gives unbounded chamber numbers. This proves existence for all retained compact diagrams, including nonstrings, but supplies no small-order bound. Rank two is expressly excluded: a finite quotient of the infinite dihedral group changes its infinity bond.

The earlier constructions remain attributed to their authors. Corn–Singerman [9] provide the regular-hypermap triangle-rotation presentation; Monson–Schulte [10] the modular $\{3, 5, 3\}$ examples; Breda–Jones–Schulte [11] the $D_s^6 \rtimes [3, 5]$, $s \geq 2$, construction; Hartley–Leemans [12] the $J_1 \times J_1$ and $J_1 \times J_1 \times L_2(19)$ examples. The star-polytope construction of McMullen [13], also described by Coxeter [14], has the stated rank-four 120 facets and vertex figures and group $H_4$ of order 14400. Its deep-hole relations are retained as part of that specific construction. The rank-five deep-hole object cited on source page 13 is an infinite 5-apeirotope; that page does not explicitly call it finite. Theorem 3 proves finite existence independently and does not reproduce these particular groups or historical constructions.

## 6 The complete finite-table disposition

For a square $(a, b, c, d)$ the four cycle edges have those labels and the diagonals label two. Strings $[a, b, c]$ have three consecutive bonds, and $[5, 3_3]$ is the rank-four fork with bonds $5, 3, 3$. This fixes the ordered Coxeter presentations up to the source's permitted diagram relabelling. Every regular hypertope with such a diagram gives a normal kernel in this presentation of index equal to its number of chambers. No assumption of universal local residues enters this necessary enumeration interface.

The eight-type core [21] completely checks all 120 normal kernels through its eight respective bounds and obtains 59 positive marked kernels. Its test checks all marked subgroup intersections and all sets of objects incident with each base flag. Induction on flag size proves that this tests *all* actual flags, not only those already known to lie in chambers. It therefore pays flag extension, thinness, residual connectivity, and the full type-preserving group. The separate structural certificates in [22] prove exactly the following twenty-two source group rows. Here split actions are the actual certified actions; notation alone is not a uniqueness or isomorphism proof.

| Diagram | Certified source group | Order |
|---|---|---|
| $(3,3,3,4)$ | $2^4 : A_5 : 2$ | 1920 |
| $(3,3,3,4)$ | $(2^3 : L_2(7)) \times C_2$ | 2688 |
| $(3,3,3,4)$ | $2^5 : A_5 : 2$ | 3840 |
| $(3,3,3,4)$ | $(4 \cdot (2^4 : A_5)) : 2$ | 7680 |
| $(3,3,3,4)$ | $2^6 : A_5 : 2$ | 7680 |
| $(3,3,3,5)$ | $(A_5 \times A_5) : 2$ | 7200 |
| $(3,4,3,4)$ | $2^{1+4} : S_3 : S_3$ | 1152 |
| $(3,4,3,4)$ | $S_6 \times C_2$ | 1440 |
| $(3,4,3,4)$ | $S_6 \times S_3$ | 4320 |
| $(3,4,3,4)$ | $(2^3 : L_2(7)) \times C_2^2$ | 5376 |
| $(3,4,3,4)$ | $L_2(7) \times S_3^2$ | 6048 |
| $(3,4,3,4)$ | $2^8 : S_3 : S_3$ | 9216 |
| $[3,5,3]$ | $L_2(16) : 2$ | 8160 |
| $[4,3,5]$ | $2^5 : A_5 : 2$ | 3840 |
| $[4,3,5]$ | $(2^4 : A_5 : 2) \times C_2$ | 3840 |
| $[4,3,5]$ | $(2^5 : A_5 : 2) \times C_2$ | 7680 |
| $[4,3,5]$ | $2^5 : A_5 : 2 : 2$ | 7680 |
| $[5,3,5]$ | $(A_5 \times A_5) : 2$ | 7200 |
| $[5,3,5]$ | $L_2(16) : 2$ | 8160 |
| $[5,3_3]$ | $(2^5 : A_5) \times C_2$ | 3840 |
| $[5,3_3]$ | $2^6 : A_5$ | 3840 |
| $[5,3_3]$ | $(2^6 : A_5) \times C_2$ | 7680 |

The displayed nonsplit dot has an actual central/derived obstruction to a complement. Split and direct factors have literal certificates, including the affine $2^3 : L_2(7)$ model, extraspecial $2^{1+4}$, and the outer involution in $L_2(16) : 2$. Every row in this positive table has universal spherical local groups.

Four printed rows are excluded even under the broad convention:

| Diagram | Printed order | Complete exclusion |
|---|---|---|
| $(3,4,3,4)$ | 336 | all three exact-order kernels fail IP |
| $(3,5,3,4)$ | 2160 | the exact-order kernel fails IP |
| $(3,5,3,5)$ | 3840 | all eleven exact-order kernels fail IP |
| $(3,5,3,5)$ | 9720 | the exact-order kernel fails IP |

The first two follow from [21]. The complete ninth-square search through 3840 [24] returns 25 kernels and 23 markings after the necessary rank-two divisibility filter. Twenty-one fail the intersection property. The order-960 candidate satisfies it but has a nonextendible three-type flag. The sole positive has order 1920 and all four local orders 60. Its explicit permutation tuple and full incidence proof are supplied; its global abstract structural name is not asserted.

For order 9720, the exact-divisor method [25] preserves every normal overgroup of a sought kernel: its index must divide 9720. The six returned indices are 1, 2, 60, 120, 4860 and 9720, each once. Its sole exact-order quotient fails twelve marked intersections. An independent integer closure exhibits the same nonidentity central involution in all four maximal parabolics. This excludes that row without repeating the timed-out unrestricted search through 9720, and makes no completeness claim at unrelated indices.

The complete bounded minima, including all source-equivalent markings, are:

| Diagram | Complete bound | Broad minimum | Universal minimum |
|---|---|---|---|
| $(3,3,3,4)$ | 7680 | 168 | 1920 |
| $(3,3,3,5)$ | 7200 | 2520 | 7200 |
| $(3,4,3,4)$ | 9216 | 168 | 1152 |
| $(3,5,3,4)$ | 2160 | 1080 | $> 2160$ |
| $(3,5,3,5)$ | 3840 | 1920 | $> 3840$ |
| $[3,5,3]$ | 8160 | 660 | 8160 |
| $[4,3,5]$ | 7680 | 1920 | 3840 |
| $[5,3,5]$ | 8160 | 3420 | 7200 |
| $[5,3_3]$ | 7680 | 960 | 3840 |

The broad minima are all attained. An inequality in the last column means only exclusion through that bound; no attained universal minimum beyond it is claimed. Source Table 3's assertion that each first row is smallest is therefore replaced by this explicit two-convention table. For example the product-size obstruction $48^2/6 = 384 > 336$ applies only to universal $B_3$ residues; the full three-kernel test is what excludes 336 without that hypothesis. Kernel inventories retain all multiplicities and are compared by exact ordered-generator isomorphisms, not compact permutation byte equality or abstract group names.

These finite exhaustions retain the correctness and completeness contract of GAP/LINS, its default normal-subgroup traversal, and its fixed tables of characteristically simple groups, multiplier primes and socle overgroups. The installed maximum index $10^7$ exceeds every bound used. The acquired Coxeter seed and module-interface proofs [23] and divisor proof state the precise retained interfaces. Literal closures and coset readers independently verify returned markings, but do not constitute a formal verification of the normal-subgroup algorithm or its classification tables.

## 7 Source disposition, premises and attribution

The clause ledger contains 109 entries: the original 83 stable source clauses, the 21 explicit Table 2 rows, and five scope clauses. It routes all numbered statements, both examples, all ten Table 1 rows, all affine families, all twenty-six Table 3 rows and nine minimum assertions. Corrections are local to their stated assertions: Theorem 3.2; local injectivity in the broad reading of Theorem 3.4; $E_6$'s order; the overbroad Coxeter-element half-power sentence; sufficiency of affine parabolic avoidance; and the four finite-table rows and consequent minimum claims. The source's rank-five small search remains an inconclusive historical search, not a nonexistence theorem. No formal named conjecture is disposed of here.

The indispensable retained inputs are the classical Coxeter complex and parabolic/chamber facts; faithful geometric representations and affine alcove stabilizers; finite/affine/compact diagram classifications; Maxwell's finite normal-subgroup alternative; the string correspondence; the named toroidal classifications; the cited earlier special constructions; and the exact finite software/classification-table contracts. Their roles and source citations are explicit in the ledger. In particular, neither a new all-rank affine lattice classification nor independent reconstructions of every historical example are claimed. The source's abstract must be read with its Section 5 review-and-announcement boundary.

The assembly checker verifies the exact acquired manuscripts, certificates, reviews and their declared dependencies, the source-table joins, and the native document. Small independent controls exercise the new direct-sum, coset-kernel and $D_4$ joins and reject damaged ledgers. It does not rerun an acquired group census or exceptional weight-orbit enumeration. Mathematical validity of the new joins requires the separate reading of this proof; hash identity is only an acquisition check.

The classifications and source questions belong to the cited authors. Component authorship and distinct review roles are preserved in their records. The present author previously

contributed the eight-type finite core, named-group certificates and Coxeter interfaces, and independently reviewed the separate divisor, ninth-square and cubic applications. The present synthesis and elementary joins require their own distinct review. No claim of historical novelty, new formalization, new question credit or measured compression follows from this assembly.

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[23] Human Compression project, complete reviewed Coxeter seed and LINS composition interfaces, L033 method component.

[24] Human Compression project, complete square $(3, 5, 3, 5)$ quotients through order 3840, L033 ninth-square component.

[25] Human Compression project, exact-divisor Coxeter quotient search and the order-9720 exclusion, L033 divisor component.

[26] Human Compression project, infinite cubic example and its exact dual-cell interface, L033 Example 2.5 component.

# Residue stabilizers and distinguished generators

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This manuscript supplies the general arguments for the intake of Fernandes–Leemans–Weiss, *Highly symmetric hypertopes* (2016). Its central observation is that a stabilizer of a flag can be read from paths of chambers in its residue. Ordinary paths give the distinguished involutions of a regular hypertope; even paths give the distinguished generators of a chiral hypertope. Intersections then follow from intersections of pointwise stabilizers. We use the already acquired propagation, extension-fiber and reflection theorems in the chamber manuscript. Their proofs are dependencies, not new assertions of the present application. Finite examples, numerical tables and the toroidal families are separate components.

## 1 Conventions and acquired geometric arguments

Let $I$ be a finite type set. An incidence geometry has one object of each type in every chamber, and every flag extends to a chamber. Distinct objects of the same type are not incident. The residue of a flag $F$ consists of the objects of the remaining types incident with every object of $F$. Residual connectedness means incidence connectedness of every residue of rank at least two, including the empty-flag residue. Thinness means that every panel is contained in exactly two chambers.

For a thin geometry write $C^i$ for the unique chamber different from $C$ on the panel obtained by deleting its type-$i$ object. Superscripts are applied in their written order: $C^{ij} = (C^i)^j$. All group actions in this manuscript are *left* actions; thus $(gh)C = g(hC)$. An automorphism preserves types unless stated otherwise. It commutes with every operation $C \mapsto C^i$.

The acquired propagation and reconstruction lemma has three conclusions:

1. Chambers containing a fixed flag are connected by adjacency steps which retain that flag.
2. In a thin residually connected geometry, an automorphism fixing a chamber is the identity.
3. A permutation of chambers preserving every adjacency color comes from a unique automorphism of the incidence geometry.

The proof of the first conclusion is an induction on the rank of the residue. It is important to apply the induction to residues known to be connected. Mere incidence connectedness of the whole geometry does not supply that hypothesis. The converse needed for Proposition 2.1 of the source is elementary: a chamber path in a residue of rank at least two gives an incidence path between any two of its objects. Consecutive chambers share at least one object, and each object lies in a chamber. Thus residual connectedness and strong chamber connectedness are equivalent under the source's hypotheses. This corrects an intermediate step in its proof without changing the stated proposition.

Chamber transitivity implies transitivity on flags of every fixed type: extend both flags to chambers and carry one chamber to the other. Conversely chamber transitivity is a special case of flag transitivity. This is Proposition 2.2 of the source.

## 2 The residue stabilizer identity

Fix a chamber $C = \{c_i : i \in I\}$ and put

$$F_{I\setminus S} = \{c_i : i \notin S\} \qquad (S \subseteq I).$$

Regularity means transitivity of the full automorphism group on chambers; by propagation its action is free as well. The same conclusions below hold for any chamber-transitive group of actual automorphisms, which then equals the full group by freeness.

**Theorem 2.1.** *Let $G$ act regularly on the chambers of a hypertope. There is a unique $\rho_i \in G$ with $\rho_i C = C^i$ for each $i \in I$. These elements are distinct nonidentity involutions. For every $S \subseteq I$,*

$$\langle \rho_i : i \in S\rangle = \mathrm{Stab}_G(F_{I\setminus S}). \tag{1}$$

*In particular they generate $G$ and satisfy the full intersection property*

$$\langle \rho_i : i \in S\rangle \cap \langle \rho_i : i \in T\rangle = \langle \rho_i : i \in S \cap T\rangle.$$

*Proof.* Existence and uniqueness of $\rho_i$ follow from regularity. Since $\rho_i(C^i) = (\rho_i C)^i = C$, its square fixes $C$, so $\rho_i^2 = 1$. It is nonidentity because $C^i \neq C$. The chambers $C^i, C^j$ are distinct when $i \neq j$, so the generators are distinct. Each $\rho_i$ fixes every $c_j$ with $j \neq i$. This proves inclusion from left to right in (1).

Let $g$ fix $F_{I\setminus S}$. The chambers $C$ and $gC$ contain this flag. Propagation inside its residue gives a path from $C$ to $gC$ whose colors all belong to $S$. If its colors are $i_1, \ldots, i_m$, its successive chambers are

$$C, \quad \rho_{i_1}C, \quad \rho_{i_1}\rho_{i_2}C, \quad \ldots, \quad \rho_{i_1}\cdots\rho_{i_m}C.$$

Indeed adjacency commutes with every group element. The last product and $g$ have the same image of $C$, and freeness makes them equal. This also works for $S = \varnothing$, when the stabilizer is trivial. It proves (1). Taking $S = I$ proves generation. Finally, to fix both complementary flags is exactly to fix their union, which is the complementary flag of $S \cap T$. Applying the identity three times gives the claimed intersection. □

This proves source Theorem 4.1 without an induction that presumes the identification of the residue group. Its rank-two Coxeter quotient interpretation follows as well: put $m_{ij} = o(\rho_i\rho_j)$, allowing $\infty$. The Coxeter presentation with these labels maps onto $G$. The rank-two subgroups are dihedral of order $2m_{ij}$ (infinite when the label is infinite), so the map is injective on each such subgroup.

For the chiral case, $G = \mathrm{Aut}_I(\Gamma)$ has exactly two chamber orbits, and adjacent chambers lie in different orbits. Call the orbit of $C$ positive. Every two-step chamber $C^{ij}$ is positive. The action is free by propagation, so there is a unique element taking $C$ to $C^{ij}$. Fix $0 \in I$, put $\alpha_0 = 1$, and for $i \neq 0$ define $\alpha_i C = C^{0i}$.

**Theorem 2.2.** *For a chiral hypertope of rank at least three set*

$$\alpha_{ij} = \alpha_i^{-1}\alpha_j, \qquad H_S = \langle \alpha_{ij} : i, j \in S\rangle \quad (S \subseteq I).$$

*Then $\alpha_{ij}C = C^{ij}$ and*

$$H_S = \mathrm{Stab}_G(F_{I\setminus S}). \tag{2}$$

*Consequently $G = \langle \alpha_i : i \neq 0\rangle$ and $H_S \cap H_T = H_{S\cap T}$ for all subsets $S, T$. The set of nonidentity distinguished generators $\{\alpha_i : i \neq 0\}$ is independent.*

*Proof.* First $\alpha_i^{-1}C = C^{i0}$, since applying $\alpha_i$ to the latter chamber gives $C^{0ii0} = C$. Therefore

$$\alpha_i^{-1}\alpha_j C = \alpha_i^{-1}(C^{0j}) = (C^{i0})^{0j} = C^{ij}.$$

The same identity includes $i = 0$, $j = 0$ and $i = j$. In particular $\alpha_{ij}$ fixes every object of $C$ outside types $i, j$, so $H_S$ is contained in the stabilizer in (2).

For the reverse inclusion let $g$ fix the complementary flag. Propagation gives a path from $C$ to $gC$ using colors in $S$. Every adjacency changes the chamber orbit. Since both endpoints are positive, the path has even length, say with colors $i_1, j_1, \ldots, i_m, j_m$. After its first pair the chamber is $\alpha_{i_1 j_1} C$. Equivariance then shows successively that its last chamber is $\alpha_{i_1 j_1} \cdots \alpha_{i_m j_m} C$. Freeness makes this product equal to $g$.

This argument includes singleton and empty $S$. Their group $H_S$ is trivial; equivalently a group element stabilizing a panel cannot exchange its two chambers because those chambers have different colors. We have proved the stabilizer identity for every $S$, rather than assumed that a residue has at most two orbits. Generation follows on taking $S = I$; intersections follow by taking the union of the two fixed flags.

The chamber $C^{0i}$ differs from $C$ for $i \neq 0$: equality would imply $C^0 = C^i$, which is impossible for different types. Hence $\alpha_i \neq 1$. The subgroup generated by the other $\alpha_j$ is $H_{I\setminus\{i\}}$, whereas $\alpha_i \in H_{\{0,i\}}$. Their intersection is $H_{\{0\}} = 1$. Thus no $\alpha_i$ lies in the subgroup generated by the others. □

The intersection conclusion contains the source's condition $IC^+$, which is stated only for $|S|, |T| \geq 2$. This proves Theorem 7.1 and Corollary 7.2. Ranks at most two have no chiral hypertopes: a connected thin rank-two incidence graph is a cycle or a doubly infinite path, whose type-preserving automorphisms are transitive on its edges; the rank-one case is the two-object geometry.

**Corollary 2.3.** *The chiral group is transitive on flags of every proper type set. In particular every proper truncation is flag-transitive.*

*Proof.* Let $F$ be a flag of proper type $J$ and extend it to a chamber $D$. If $D$ is negative, choose $i \notin J$ and replace it by $D^i$; this retains $F$ and makes the chamber positive. Some $g \in G$ takes $C$ to the resulting positive chamber and therefore takes its $J$-subflag to $F$. A flag in a truncation extends to a chamber of that truncation by extending it in the whole geometry. The same transitivity argument applies there. □

This includes Proposition 8.1 of the source. It asserts transitivity, not thinness or regularity of those truncations.

## 3 Cosets and the converse construction

For subgroups $P_i \leq G$, the objects of type $i$ are left cosets $gP_i$, with incidence given by nonempty intersection. This always defines an incidence system and has the base chamber $\{P_i : i \in I\}$. It need not have the flag-extension property in rank at least four. Left multiplication is transitive on objects of each type and on incident pairs: take a common representative of the two cosets. These are the precise assertions used from source Proposition 3.1.

The kernel of the action is

$$K = \bigcap_{i \in I} \bigcap_{g \in G} g P_i g^{-1}.$$

It is contained in every $P_i$. Sending $gP_i$ to $(gK)(P_i/K)$ is a bijection on each type. It preserves incidence in both directions, because a representative of an intersecting quotient coset can be lifted and the discrepancy is in $K \leq P_i$. Thus it gives the isomorphism of source Theorem 3.3. A claim that $G$ itself acts faithfully before this quotient would be false.

If a group is transitive on objects and incident pairs of each type, its geometry is reconstructed by these cosets of the stabilizers of a base chamber. Indeed $gP_i \mapsto gc_i$ is a well-defined bijection; two image objects are incident precisely when the pair is a translate of the base pair, which is equivalent to the two cosets intersecting. This applies to regular hypertopes and, by Corollary 2.3, to chiral hypertopes of rank at least three. The residue stabilizer identities identify their $P_i$ with the prescribed marked subgroups.

For a regular faithful action the Borel subgroup $B = \bigcap_i P_i$ is the chamber stabilizer and hence is trivial. The stabilizer of a panel has exactly two elements, since it acts transitively and freely on the two containing chambers. These observations prove Lemma 3.4 and its stated consequence. For a nonfaithful chamber-transitive coset action on a hypertope, the corresponding conclusion is $B = K$ and a panel stabilizer has quotient of order two by $K$; passage to the faithful quotient is essential.

**Theorem 3.1.** *Let $(G, \{\rho_i : i \in I\})$ be a C-group and put $G_S = \langle \rho_i : i \in S \rangle$ and $P_i = G_{I\setminus\{i\}}$. If its coset incidence system is flag-transitive under $G$, it is a regular hypertope. Its full automorphism group is $G$.*

*Proof.* Every flag is a translate of the corresponding base flag, so extends to a chamber. Iterated use of the C-group intersection property gives

$$\bigcap_{j \notin S} P_j = G_S \qquad (S \subseteq I),$$

including the conventions for empty intersections. The base chamber stabilizer is $G_\varnothing = 1$, so the action is faithful and free on chambers. The stabilizer of the complementary base flag acts transitively on the chambers containing that flag: a group element taking the base chamber to any such chamber must fix every object of the flag. For a panel its stabilizer is $G_{\{i\}} = \{1, \rho_i\}$; hence the geometry is thin.

More generally its residue chambers are the chambers $gC$ with $g \in G_S$. A word in the $\rho_i$, $i \in S$, gives an adjacency path between $C$ and $gC$ wholly inside that residue. Thus all residues are chamber-connected and therefore incidence-connected in rank at least two. The geometry is a hypertope. Its full automorphism group acts freely on chambers, whereas $G$ already acts transitively, so the two groups coincide. □

This proves source Theorem 4.6. It does not assert flag transitivity from the intersection property alone.

# 4 The classical coset criteria

The following elementary arguments record exactly which group conditions are used. Their origins remain Tits, Dehon and Buekenhout–Hermand as attributed in the source; they are not new claims of priority.

**Proposition 4.1.** *Suppose the coset incidence system is flag-transitive. Put $P_J = \bigcap_{j\in J} P_j$, with $P_\varnothing = G$. It is residually connected if and only if*

$$P_J = \langle P_{J\cup\{k\}} : k \in I \setminus J \rangle \qquad (|I \setminus J| \geq 2).$$

*Proof.* Fix the base flag of type $J$ and write $H = P_J$. Every incident object of type $k \notin J$ has the form $hP_k$ with $h \in H$, by flag transitivity on $J \cup \{k\}$. Two such representatives give the same object precisely when they give the same left coset of $H \cap P_k$ in $H$. If two objects of the residue are incident, flag transitivity on their union with the base flag supplies a common representative in $H$. Consequently the residue is exactly the coset incidence system of $H$ with subgroups $H \cap P_k$.

In a coset incidence graph of rank at least two, the component containing the base flag is invariant under each of its vertex stabilizers: an automorphism fixing a vertex preserves that vertex's component. It therefore contains every vertex represented by an element of the group $L$ generated by those stabilizers. Conversely, since each stabilizer is contained in $L$, assigning the label $hL$ to a vertex represented by $h$ is well-defined and constant along edges. The graph is connected if and only if $L = H$. Apply this to every $J$. □

Without flag transitivity, a residue can have additional incidence edges; the equality with a coset graph must not be assumed.

**Proposition 4.2.** *For each nonempty $J \subseteq I$ choose $a(J) \in J$. The coset system is flag-transitive if and only if, for every $|J| \geq 3$,*

$$\bigcap_{j \in J \setminus \{a(J)\}} P_j P_{a(J)} = \left( \bigcap_{j \in J \setminus \{a(J)\}} P_j \right) P_{a(J)}.$$

*Proof.* The acquired extension-fiber theorem identifies orbits on flags of type $T \cup \{k\}$, assuming transitivity on type $T$, with

$$P_T \backslash \left( \bigcap_{j \in T} P_j P_k \right) / P_k.$$

There is one orbit precisely when the middle set is $P_T P_k$. Transitivity on flags of size at most two always holds. Inducting on $|J|$, with $T = J \setminus \{a(J)\}$, proves sufficiency. Necessity is the same orbit calculation at every step. Every flag then translates to a base flag and extends, so no missing geometry hypothesis is hidden in this test. □

**Lemma 4.3.** *For subgroups $A, B, D$ the following are equivalent:*

$$AB \cap AD = A(B \cap D), \tag{i}$$

$$(A \cap B)(A \cap D) = A \cap BD, \tag{ii}$$

*and (iii): three right cosets of $A, B, D$ that meet pairwise have a common point.*

*Proof.* Suppose (i), and let $x = bd \in A \cap BD$. Then $b = xd^{-1}$ belongs to $AB \cap AD$, so $b = ae$ with $a \in A$ and $e \in B \cap D$. Since $b, e \in B$, we have $a \in A \cap B$; and $ed = a^{-1}x$ belongs to $A \cap D$. This proves the nontrivial inclusion of (ii).

Conversely let $x = ab = cd \in AB \cap AD$, with $a, c \in A$, $b \in B$, $d \in D$. Inverting equality (ii) also gives $A \cap DB = (A \cap D)(A \cap B)$. Thus $c^{-1}a = db^{-1} = uv$, where $u \in A \cap D$ and $v \in A \cap B$. It follows that $x = cu(vb)$, with $vb = u^{-1}d \in B \cap D$. This gives (i); the opposite inclusions in both equalities are automatic.

For the coset assertion, translate a common point of the $B$- and $D$-cosets to the identity by right multiplication. The three cosets then have the form $Ak, B, D$. They meet pairwise precisely when $k \in AB \cap AD$. They have a common point precisely when $k \in A(B \cap D)$. This is exactly (i). □

This is source Lemma 4.2. Inverting cosets gives the equivalent statement for the left-coset convention used elsewhere here.

**Corollary 4.4.** *For a rank-three C-group the standard coset geometry is thin if and only if the designated group acts regularly. In either case it is a regular hypertope.*

*Proof.* Any rank-three coset system is a geometry: an incident pair has a common representative, which supplies an object of the third type. If the group is chamber-transitive, Theorem 3.1 applies.

Now suppose it is thin. Normalize any incident pair to $P_i, P_j$, and let $k$ be the remaining type. There are two distinct completions $P_k$ and $\rho_k P_k$: the element $\rho_k$ belongs to $P_i \cap P_j$ and not to $P_k$, by the intersection property. Both completed chambers are translates of the base chamber. Thinness allows no other completion. Hence every chamber is a translate of the base chamber, and the theorem applies. No inference of freeness on all chambers from the base stabilizer alone is used. □

## 5 Rotation subgroups and strings

**Proposition 5.1.** *Let $(G, \{\rho_i : i \in I\})$ be a C-group with $0 \in I$. Set $\alpha_0 = 1$, $\alpha_i = \rho_0\rho_i$ for $i \neq 0$ and $H = \langle \alpha_i : i \neq 0 \rangle$. These generators are independent. The index $[G : H]$ is one or two. If it is two, then*

$$\langle \alpha_i^{-1}\alpha_j : i, j \in S \rangle = G_S \cap H \qquad (S \subseteq I),$$

*so the rotation generators satisfy $IC^+$.*

*Proof.* Conjugation by $\rho_0$ inverts every $\alpha_i$, and $\rho_i = \rho_0\alpha_i$. Thus $H \lhd G$ and $G/H$ is generated by the image of $\rho_0$, whose order is at most two. If $\alpha_i$ lay in the subgroup generated by the other $\alpha_j$, it would belong to $G_{I\setminus\{i\}}$. Multiplication by $\rho_0$ would put $\rho_i$ in that subgroup, contradicting the intersection property. This proves independence regardless of the index.

If the index is two, all $\rho_i$ have the same nontrivial image in $G/H$. Consequently a word in the $\rho_i$, $i \in S$, represents an element of $H$ exactly when its length is even. Every such word is a product of pairs $\rho_i\rho_j = \alpha_i^{-1}\alpha_j$ with $i, j \in S$. This proves the displayed identity. Intersecting two such identities and using $G_S \cap G_T = G_{S\cap T}$ proves $IC^+$. □

These are Propositions 6.1 and 6.2. The index-two hypothesis in the last assertion cannot be discarded; the source's Example 6.3 is checked in the finite component.

For source Theorem 5.1 we retain the standard string C-group correspondence with abstract regular polytopes, in its exact marked form: the faces of type $i$ are the cosets of $G_{I\setminus\{i\}}$, its flag action is regular, and its rank-two sections have polygon sizes $o(\rho_i\rho_j)$. This is the classical Aschbacher–Schulte input cited in that theorem and the McMullen–Schulte correspondence already retained by the programme. It yields flag transitivity, after which Theorem 3.1 gives the hypertope statement. This input is not counted as a new proof here.

Conversely, for a regular hypertope the rank-two residue on types $i, j$ has group $G_{\{i,j\}}$ by Theorem 2.1. Its alternating chamber cycle has length $2o(\rho_i\rho_j)$, or is infinite if that order is infinite. A missing edge in its diagram means this residue is a digon, so the product order is two. If the diagram is a string, every pair of nonconsecutive types is missing, and the distinguished C-group is therefore a string C-group. This proves source Theorem 5.2.

## 6 The two chamber orbits in Construction 8.1

Let $I = \{0, \ldots, r-1\}$ with $r \geq 3$. Suppose $G = \langle \alpha_1, \ldots, \alpha_{r-1} \rangle$ is a $C^+$-group with its independent nonidentity generators. Set $\alpha_0 = 1$ and $H_S = \langle \alpha_i^{-1}\alpha_j : i, j \in S \rangle$. The parabolics of Construction 8.1 are

$$P_i = H_{I\setminus\{i\}} \quad (i \in I).$$

For $i > 0$ this is $\langle \alpha_j : j \neq 0, i \rangle$; for $i = 0$ it is $\langle \alpha_1^{-1}\alpha_j : 2 \leq j < r \rangle$, by telescoping the pairwise differences. We continue to assume, as the source theorem does, that the associated coset system is a hypertope and that $G$ has *exactly two* orbits on its chambers.

Iterating $IC^+$ gives trivial intersections of all parabolics but one: the common support is a singleton and its $H$-group is trivial. The iteration uses $IC^+$ while both supports have size at least two; as soon as a singleton is reached the intersection is already trivial. The base chamber stabilizer is trivial as well. Thus this is a faithful action, and propagation makes it free on every chamber.

Write $C = \{P_i : i \in I\}$. The coset $\alpha_1 P_0$ contains every $\alpha_j$, $j > 0$. For each $i > 0$ choose $j > 0$ different from $i$; then $\alpha_j \in P_i$, so replacing $P_0$ by $\alpha_1 P_0$ gives a chamber. It differs from $C$: otherwise $\alpha_1 \in H_{I\setminus\{0\}} \cap H_{\{0,1\}} = H_{\{1\}} = 1$. By thinness it is $C^0$; denote it by $C_0$.

For $i > 0$, the coset $\alpha_i^{-1} P_i$ is different from $P_i$ by independence. It meets $P_k$ for $k > 0$, $k \neq i$, at $\alpha_i^{-1}$. It meets $P_0$ at $\alpha_i^{-1}\alpha_j$, where $j > 0$, $j \neq i$. Replacing $P_i$ by this coset therefore gives the chamber $C^i$. On multiplying it by $\alpha_i$, every positive-type object becomes the corresponding object of $C_0$, while $\alpha_i P_0 = \alpha_1 P_0$. Hence

$$C^i = \alpha_i^{-1} C_0 \qquad (i \in I).$$

Here $i = 0$ follows from $\alpha_0 = 1$.

No element of $G$ takes $C$ to any $C^i$, since it would fix the base panel and its stabilizer is trivial. The same holds at any translate of $C$. Also every neighbor of a translate of $C_0$ is a translate of $C$: this follows from $C^i = \alpha_i^{-1} C_0$ and involutivity of adjacency. With the exactly-two-orbits hypothesis, all chambers are therefore uniquely

$$(g, +) = gC, \qquad (g, -) = gC_0, \qquad (g, +) \sim_i (g\alpha_i^{-1}, -).$$

These are precisely the hypotheses and coordinates of the acquired two-regular-orbits theorem. It follows that the full group is $G$, and the hypertope is chiral, unless some $\theta \in \mathrm{Aut}(G)$ inverts every $\alpha_i$. If such a $\theta$ exists, the explicit color-preserving map

$$(g, +) \mapsto (\theta(g), -), \qquad (g, -) \mapsto (\theta(g), +)$$

extends to an incidence involution $F$, and the full group is $G \rtimes \langle F \rangle$, with $FgF^{-1} = \theta(g)$. This proves source Theorem 8.2 by the already acquired theorem. In this notation the adjoined chamber involution can be nontrivial even when the abstract automorphism $\theta$ is the identity.

The proof uses the two-orbits hypothesis to identify the entire chamber set. It does not establish that hypothesis for an arbitrary $C^+$ presentation. Nor does an abstract semidirect product alone establish that its extra involution acts on the proposed geometry.

# 7 Scope, attribution and source questions

Theorems 2.1 and 2.2 replace the two separate intersection arguments by the same residue-stabilizer identity. The converse construction and the rotation argument are then short deductions. The geometric propagation lemma, extension-fiber theorem and explicit reflection theorem are acquired project proofs, with their earlier attributions unchanged. The subgroup criteria remain the classical results attributed by the source to Tits, Dehon and Buekenhout–Hermand. The string correspondence remains an explicit retained theorem, not an inferred consequence of an order calculation. No claim of historical priority for the stabilizer observation is made.

The programme already has reviewed negative answers to source Problems 9.3 and 9.4 under the standard parabolic construction. Their precise statements are retained here. Thinness and chamber connectedness force the standard C-group orbit to contain all chambers, excluding the example requested in 9.3. Every rank-three standard coset system on three generators is residually connected, excluding the example requested in 9.4. The latter argument only needs a connected *spanning* coset graph in each residue and does not assert an equality of residue incidence relations without flag transitivity. These are acquisitions of earlier answers, not two new solutions generated by this intake.

Problems 9.1 and 9.2 and the conjectured completeness of the known chiral list remain separately registered questions. This component supplies no classification of all locally toroidal cases. It grants no whole-original credit before the finite and family components and the final source map are reviewed. Existing Lean declarations retain exactly their previous algebraic and action scope; no new formalization or compression measurement is asserted here.

## References


[1] M. E. Fernandes, D. Leemans and A. Ivić Weiss, *Highly symmetric hypertopes*, Aequationes Mathematicae 90 (2016), 1045–1067. Fixed accepted author manuscript, 22 pages.

[2] Human Compression project, *Chambers and marked covers*, topic-chamber/v2, sections on normalization, propagation and reconstruction, and two regular orbits. The accepted version and review are acquired by their exact artifact bindings.

[3] P. McMullen and E. Schulte, *Abstract Regular Polytopes*, Cambridge University Press, 2002, Section 2E. The marked regular-polytope correspondence is retained, as in the existing programme.

# Kernels and intersections in quotients of hypertopes

Human Compression project

**Abstract**

Injectivity on the maximal parabolic subgroups does not by itself preserve the intersection property under a quotient. We give an affine example in which every maximal parabolic embeds and the quotient geometry is flag transitive, but has only one chamber. We then state and prove the precise additional condition on products of parabolic subgroups. This corrects the quotient assertion in Theorem 3.2 of the fixed manuscript *An exploration of locally spherical regular hypertopes*. It does not dispose of that manuscript's classification results.

## 1 The direction of a quotient criterion

Let $I$ be a finite set, and let $U = \langle s_i : i \in I \rangle$ be generated by involutions. Write $U_A = \langle s_i : i \in A \rangle$ for $A \subseteq I$ and $U_i = U_{I\setminus\{i\}}$. The marking is a C-group marking when

$$U_A \cap U_B = U_{A\cap B} \qquad (A, B \subseteq I).$$

All subgroups in this note refer to the specified marking.

The source's Theorem 3.1 is a valid upward criterion: if a marked epimorphism $G \longrightarrow H$ is injective on each maximal parabolic of $G$, and $H$ is a C-group, then $G$ is a C-group. Indeed, its proper parabolics inherit the property from their isomorphic images in $H$. An element of $G_i \cap G_j$ maps into $H_{I\setminus\{i,j\}}$; lift it within $G_{I\setminus\{i,j\}}$ and use injectivity in $G_i$. The maximal-parabolic criterion proved below completes the argument.

The proof of Theorem 3.2 instead uses a map $U/N \longrightarrow U$ which takes $s_iN$ to $s_i$. Local injectivity does not make this map well defined: an additional relation in the quotient need not hold in $U$. The issue is not confined to that proof. The following example satisfies the stated hypotheses and contradicts its conclusion.

## 2 An affine example

Equip $\mathbb{R}^2$, in coordinates $(a, b)$, with the inner product whose matrix is $\left(\begin{smallmatrix} 2 & -1 \\ -1 & 2 \end{smallmatrix}\right)$. Use homogeneous coordinates for the affine maps

$$s_0 = \begin{pmatrix} 0 & -1 & 1 \\ -1 & 0 & 1 \\ 0 & 0 & 1 \end{pmatrix}, \quad s_1 = \begin{pmatrix} -1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad s_2 = \begin{pmatrix} 1 & 0 & 0 \\ 1 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

They are reflections in $a+b = 1$, $2a-b = 0$, and $-a+2b = 0$, respectively. Put $U = \langle s_0, s_1, s_2 \rangle$ and $W = \langle s_1, s_2 \rangle$. The latter is the group of six linear symmetries generated by the last two reflections. If $t_v$ denotes translation by $v$, then

$$s_\theta = s_1 s_2 s_1 = \begin{pmatrix} 0 & -1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad s_0 = t_{(1,1)} s_\theta.$$

Conjugating $t_{(1,1)}$ by $s_1$ gives $t_{(0,1)}$, and their quotient gives $t_{(1,0)}$. Every generator has integral translation part and linear part in $W$. Consequently

$$U = \mathbb{Z}^2 \rtimes W, \qquad N = \{t_v : v \in \mathbb{Z}^2\} \lhd U, \qquad U/N \cong W \cong S_3. \tag{1}$$

We verify that $U$ is the full type-preserving group of a regular hypertope, so no converse about arbitrary marked groups is being assumed. The lines

$$2a - b \in \mathbb{Z}, \qquad -a + 2b \in \mathbb{Z}, \qquad a + b \in \mathbb{Z}$$

tile the plane by equilateral triangles. A fundamental triangle is

$$\Delta = \{2a - b \geq 0,\ -a + 2b \geq 0,\ a + b \leq 1\},$$

with vertices 0, $\lambda_1 = (2/3, 1/3)$, and $\lambda_2 = (1/3, 2/3)$. The vertex set is the disjoint union of $\mathbb{Z}^2$, $\lambda_1 + \mathbb{Z}^2$, and $\lambda_2 + \mathbb{Z}^2$. Give these three sets different types. Each generator preserves each set, as follows directly from the displayed matrices. Declare vertices incident when they coincide or are joined by a tiling edge.

Each edge belongs to two triangles; each vertex link is a hexagon; every three pairwise adjacent vertices form a tiling triangle. It follows that this incidence structure is a geometry, is thin, and is residually connected. Here thinness concerns extensions of an edge by a vertex of the third type, and residual connectivity concerns the whole tiling graph and its hexagonal vertex links.

The group $U$ is transitive on triangles. To see this, cross a path of adjacent triangles starting at $\Delta$. Once the current triangle is $u\Delta$, reflection in one of its sides is $us_iu^{-1}$; the next triangle is therefore $us_i\Delta$. The adjacency graph is connected, so induction reaches every triangle. The action is free on triangles. An element stabilizing $\Delta$ preserves types and hence fixes each of its three noncollinear vertices, which forces an affine map to be the identity. Finally, a type-preserving automorphism fixing one chamber fixes each neighboring chamber, since an edge has exactly two extensions. Propagation across the connected chamber graph fixes the whole geometry. Thus $U$ is its full type-preserving automorphism group, acting regularly on chambers. Denote this regular hypertope by $\mathcal{U}$.

**Proposition 1.** *For each $i \in \{0, 1, 2\}$, the subgroup $U_i$ has order six and $N \cap U_i = 1$. Nevertheless the coset geometry with group $U/N$ and maximal marked parabolics $\pi(U_i)$ has one face of each type. It is flag transitive and is not a hypertope.*

*Proof.* Any pair of the three reflection matrices consists of involutions whose product has order three. A group generated by two such involutions is a quotient of the dihedral group of order six: every word reduces to $(s_is_j)^k$ or $(s_is_j)^ks_i$, $0 \leq k < 3$. The linear parts of each pair are distinct transpositions generating the six-element group $W$. Hence each pair generates exactly six elements and the linear-part map is injective on that subgroup. This proves $N \cap U_i = 1$ and $\pi(U_i) = W$ for all $i$.

Each coset set $W/\pi(U_i)$ is therefore a singleton. With types kept distinct, all three faces are incident and there is exactly one chamber. Every flag extends to it and $W$ is transitive on flags of each prescribed type. Every residue of rank one has one face, so the geometry is not thin. The action of $W$ has kernel $W$; faithfulness was not a hypothesis of the source theorem. The orbit quotient $\mathcal{U}/N$ is exactly the same geometry, since translations are transitive on vertices of each type. □

## 3 The additional kernel condition

**Proposition 2.** *Suppose $G = \langle r_i : i \in I\rangle$, with $|I| \geq 2$, and all maximal parabolics $G_i$ are C-groups on their indicated generators. Then $G$ is a C-group if and only if $G_i \cap G_j = G_{I\setminus\{i,j\}}$ for all distinct $i, j$.*

*Proof.* Necessity follows from the definition. For sufficiency, let $x \in G_A \cap G_B$. If either index set is $I$, the conclusion is immediate. Otherwise choose $i \notin A$ and $j \notin B$. If $i = j$, the desired intersection follows inside the C-group $G_i$. If $i \neq j$, the assumed maximal intersection places $x$ in $G_{I\setminus\{i,j\}}$. Intersecting with $G_A$ inside $G_i$ places $x$ in $G_{A\setminus\{j\}}$. Both this subgroup and $G_B$ lie in $G_j$, whose intersection property now gives $x \in G_{A\cap B}$. The reverse inclusion always holds. □

**Theorem 3.** *Let $U = \langle s_i : i \in I\rangle$ be a C-group with $|I| \geq 2$, let $N \triangleleft U$, and assume $N \cap U_i = 1$ for every $i$. The induced marking on $U/N$ is a C-group marking if and only if*

$$N \cap U_i U_j = \{1\} \qquad (i \neq j). \tag{2}$$

*In particular, if (2) holds and the quotient coset geometry is flag transitive, it is a regular hypertope.*

*Proof.* Write $\pi : U \longrightarrow U/N$ and $H_i = \pi(U_i)$. The local injectivity assumption identifies $H_i$, with its marking, with the C-group $U_i$.

Suppose (2) holds, and let $x \in H_i \cap H_j$. Choose $a \in U_i$ and $b \in U_j$ with $\pi(a) = x = \pi(b)$. Then $ab^{-1} \in N \cap U_i U_j$, so $a = b$. As $U$ is a C-group, this common element lies in $U_i \cap U_j = U_{I\setminus\{i,j\}}$. Thus $H_i \cap H_j = \pi(U_{I\setminus\{i,j\}})$. Proposition 2 gives the full intersection property.

Conversely suppose $U/N$ is a C-group. Write an element of $N \cap U_i U_j$ as $ab^{-1}$ with $a \in U_i$, $b \in U_j$. The common image of $a$ and $b$ lies in $H_i \cap H_j = \pi(U_{I\setminus\{i,j\}})$. Choose a preimage $c$ in that smaller subgroup. Injectivity on $U_i$ gives $a = c$, and injectivity on $U_j$ gives $b = c$. Hence $ab^{-1} = 1$.

For the final assertion, apply the C-group reconstruction theorem under its flag-transitivity hypothesis. This is Theorem 2.3 of the fixed source, whose complete proof is acquired from the project's reviewed manuscript on residue stabilizers and distinguished generators. □

The missing condition can be seen explicitly in our example: $s_\theta \in U_0$, $s_0 \in U_1$, but $s_\theta s_0 = t_{(-1,-1)}$ is a nonidentity element of $N \cap U_0 U_1$. Thus the repair excludes the example for the precise reason isolated in the proof.

**Scope and verification.** The fixed source is a sixteen-page revised author manuscript. The affected assertion and its proof are on page 6. Proposition 2 is its Proposition 2.1, attributed there to earlier work, and is reproved here for completeness. The affine group and its triangular complex are standard; the explicit argument above supplies all properties needed by the example. The kernel test is an elementary quotient-intersection criterion; no claim of historical priority is made. The accompanying program checks the displayed integer matrices, all three six-element parabolics, their exact linear images, and the nonzero kernel witness. The claims concerning the infinite tiling use the written proof. This note corrects one auxiliary theorem. It is neither a solution of an open conjecture nor a complete treatment of the source's spherical, Euclidean, and hyperbolic classifications.

## References


[1] M. E. Fernandes, D. Leemans, and A. Ivić Weiss, *An exploration of locally spherical regular hypertopes*. Fixed sixteen-page revised author manuscript, catalogue L033, especially Propositions 2.1–2.2 and Theorems 2.3, 3.1–3.2.

[2] Human Compression project, *Residue stabilizers and distinguished generators*. Complete reviewed L052 general component; only the reconstruction theorem with its stated flag-transitivity hypothesis is used here.

# Spherical hypertopes from marked kernels and antipodal flags

Human Compression project

**Abstract**

We give a common argument for the spherical classification in the fixed manuscript *An exploration of locally spherical regular hypertopes*. Maxwell's normal-subgroup theorem is retained as a named input. Preservation of the marked Coxeter diagram then leaves only the universal complex and, where it exists, its antipodal quotient. A single explicit flag excludes all even $D_n$ antipodal quotients. For $E_7$ and $E_8$, the remaining geometric condition reduces to exact orbits of fundamental weights, avoiding an enumeration of the full groups. Complete orbit certificates accompany the argument.

## 1 Retained inputs and the marked quotient

We use the following standard facts about a finite Coxeter system $(W, S)$. Its geometric representation is faithful and orthogonal; a closed fundamental chamber meets each orbit in exactly one point. The stabilizer of a face is its indicated standard parabolic subgroup. The spherical Coxeter complex is a flag complex, and $W$ acts regularly on its chambers. The irreducible finite diagrams and their group orders are the usual finite Coxeter classification, also used in Table 1 of the source. These facts are inputs, not new classifications established by this manuscript.

The additional normal-subgroup input is Maxwell's Theorem 0.1, quoted as Theorem 4.1 of the fixed source: if $W$ is a finite irreducible Coxeter group and $N \triangleleft W$, then either $N = \{1, -1\}$ or $N$ is the kernel of a graph homomorphism to a finite Coxeter group. Here a graph homomorphism sends each simple generator to a simple generator or to the identity, and its images contain every simple generator of the target. The full Maxwell paper has not been independently acquired; this is an explicitly retained classification input through the fixed source, with its stated meaning of "graph homomorphism."

For a regular hypertope, the distinguished chamber automorphisms are nonidentity involutions. Their pairwise product orders are exactly the labels of its Coxeter diagram, including order two for an absent edge. The resulting marked group is a quotient of the Coxeter system with that diagram. It is a C-group, and its coset geometry is the original hypertope. These are the acquired reconstruction facts of the L052 general manuscript. In particular, distinct distinguished generators cannot become equal.

**Lemma 1.** *Let $(W, \{s_i\}_{i\in I})$ and $(W', \{t_j\}_{j\in J})$ be Coxeter systems. Suppose a graph homomorphism $f : W \to W'$ sends every $s_i$ to a nonidentity element, sends distinct $s_i$ to distinct elements, and preserves the order of every product $s_i s_k$. Then $f$ is an isomorphism of marked Coxeter systems, after a bijection of their index sets.*

*Proof.* Write $f(s_i) = t_{\sigma(i)}$. The assumptions on images make $\sigma$ injective; the definition of graph homomorphism makes it surjective. The order assumption says that the two Coxeter matrices are equal under $\sigma$. The defining presentations therefore give a homomorphism in the opposite direction, taking $t_{\sigma(i)}$ to $s_i$. Both compositions fix every generator, so both are identity homomorphisms. □

**Theorem 2.** *Subject to the stated Maxwell input, a regular hypertope whose diagram is finite and irreducible is the universal Coxeter complex or its antipodal quotient. The latter possibility requires $-1 \in W$ and requires the quotient coset geometry to be a hypertope.*

*Proof.* Let $G$ be the distinguished group of the hypertope. Its marked epimorphism $W \to G$ has kernel $N$. If Maxwell's alternative expresses $N$ as the kernel of a graph homomorphism $f : W \to W'$, then $W' \cong W/N \cong G$. Thus the simple images of $f$ are nontrivial, distinct, and preserve every original product order. Lemma 1 gives $N = 1$. The remaining alternative is precisely $N = \{1, -1\}$. Reconstruction identifies the actual hypertope with the coset geometry of the indicated quotient and marking. No assumption that its proper residues are universal has been used. □

This argument avoids a separate inspection of the normal-subgroup tables for each finite type. It also avoids using injectivity on a parabolic before knowing that the corresponding residue is universal.

## 2 Parabolics in the antipodal quotient

Let $\alpha_1, \dots, \alpha_n$ be simple roots in the faithful real representation, and choose fundamental vectors $\omega_i$ with $\langle \omega_i, \alpha_j \rangle = \delta_{ij}$. Their Gram matrix is the inverse of the simple-root Gram matrix. For an irreducible finite system,

$$c_{ij} := \langle \omega_i, \omega_j \rangle > 0 \qquad (1 \le i, j \le n). \tag{1}$$

For completeness, write the positive definite simple-root Gram matrix as $D^{1/2}(1 - A)D^{1/2}$, where $D$ is positive diagonal and $A$ has nonnegative entries and connected support. Positive definiteness implies that the largest eigenvalue of $A$ is less than one. For a symmetric nonnegative matrix every eigenvalue has absolute value at most its largest eigenvalue; this follows from $|x^T A x| \le |x|^T A |x|$. Hence $(1 - A)^{-1} = \sum_{m \ge 0} A^m$. Connectivity supplies a positive entry in some power for every pair of indices, proving (1).

Write $W_i = \langle s_j : j \ne i \rangle$. This subgroup fixes the nonzero vector $\omega_i$, so it cannot contain $-1$. Moreover,

$$-1 \notin W_i W_j \qquad (i \ne j). \tag{2}$$

Indeed, if $-1 = ab$ with $a \in W_i$ and $b \in W_j$, then $a\omega_j = -\omega_j$ and $a\omega_i = \omega_i$. Orthogonality would give $c_{ij} = -c_{ij}$, contrary to (1).

For rank at least three, the quotient by $\{1, -1\}$ is therefore injective on every proper parabolic and satisfies the exact product-kernel criterion proved in *Kernels and intersections in quotients of hypertopes*. It is a C-group on the induced generators. This conclusion does not assert flag transitivity: the latter is precisely what fails in even type $D_n$. For the string diagrams, the standard string C-group reconstruction theorem does supply a regular polytope and hence a regular hypertope.

## 3 Incidence and signs

The vertices of type $i$ in the spherical Coxeter complex can be represented by $W\omega_i$. Two vertices of distinct types are incident exactly when they belong to one chamber. We need the following precise numerical test.

**Lemma 3.** *For $v \in W\omega_j$, one has $\langle \omega_i, v \rangle \le c_{ij}$. Equality holds if and only if $v \in W_i\omega_j$, equivalently if $v$ is incident to $\omega_i$.*

*Proof.* If $v$ is not in the closed fundamental chamber, some simple root satisfies $\langle v, \alpha_l \rangle < 0$. Reflect in that root. This increases the inner product with $\eta = \sum_k \omega_k$ strictly, and does not decrease the inner product with $\omega_i$: the increase in the latter is zero unless $l = i$, in which case it is strictly positive. The orbit is finite, so repeating reaches its unique dominant representative $\omega_j$. This proves the inequality. If equality held initially, no step with $l = i$ could have occurred.

All steps therefore belong to $W_i$, and $v \in W_i\omega_j$. Conversely $W_i$ fixes $\omega_i$, so every member of this orbit gives equality. The parabolic description of the Coxeter complex identifies $W_i\omega_j$ as precisely the vertices of type $j$ incident to $\omega_i$. □

Suppose $-1 \in W$. An antipodal vertex is $[v] = \{v, -v\}$. By Lemma 3, two such vertices of distinct types are incident exactly when their chosen representatives have inner product $c_{ij}$ or $-c_{ij}$. Positivity makes the sign unambiguous.

**Lemma 4.** *Every flag in the antipodal quotient lifts to a flag in the spherical complex if and only if every incident triple has positive product of its three incidence signs. It suffices to test that there are no distinct $i, j, k$ and $v \in W\omega_k$ such that*

$$\langle \omega_i, v \rangle = c_{ik}, \qquad \langle \omega_j, v \rangle = -c_{jk}. \tag{3}$$

*Proof.* Changing the sign of one representative changes the signs of its incident pairs, and leaves the product around a triangle unchanged. If all triangle products in a flag are positive, choose a first vertex and orient every other representative to have positive inner product with it. The triangle condition makes every remaining pair positive as well. These representatives are pairwise incident in the original complex; its flag property makes them a flag and hence a subflag of a chamber. Conversely any lifted flag admits all-positive representatives.

If some triangle has negative product, orient its first two representatives to make their pair positive. They are incident in the original complex, so an element of $W$ takes them to $\omega_i, \omega_j$. Choose the sign of the third representative to make its pairing with $\omega_i$ positive. Its pairing with $\omega_j$ must then be negative, which is exactly (3). This procedure is reversible. □

When the condition holds, every quotient flag lies in the image of a chamber, and $W/\{1, -1\}$ is flag transitive. The C-group reconstruction with flag transitivity now gives the hypertope. This is stronger than merely checking that every face has the correct number of adjacent chambers.

## 4 A flag excluding every even $D_n$ quotient

Use the standard realization of $D_n$ by signed permutations with an even number of sign changes, where $n \geq 4$ is even. Its fundamental vertices include

$$e_1, \qquad w = \tfrac{1}{2}(e_1 + \cdots + e_{n-1} - e_n), \qquad v = \tfrac{1}{2}(e_1 + \cdots + e_n).$$

The central antipode belongs to this group exactly when $n$ is even. Put $x = e_n$. The vertices $v, w$ are incident, as they are in the displayed fundamental chamber. The vertices $x, v$ are incident because a coordinate permutation takes $e_1$ to $x$ and fixes $v$. Likewise $-x, w$ are incident: apply the permutation interchanging coordinates $1, n$, followed by sign changes in those two coordinates. This group element takes $e_1$ to $-e_n$ and fixes $w$.

Consequently $[v], [w], [x]$ form a flag in the antipodal incidence system. Its pairings have signs $+, +, -$, since

$$\langle v, w \rangle = (n-2)/4 > 0, \quad \langle v, x \rangle = 1/2, \quad \langle w, x \rangle = -1/2.$$

It cannot lift to a chamber. This precludes regularity with the proposed marked group: if the quotient were such a regular hypertope, its chambers would be the images of the original chambers by coset reconstruction and chamber transitivity. The negative triple contradicts that requirement. Thus no regular hypertope of type $D_n$ can have the antipodal quotient as its distinguished group. This one flag handles every even rank, without a separate chamber lower bound or induction on rank.

## 5 The two exceptional incidence calculations

For $E_7$ and $E_8$, choose the simply laced Cartan matrix $C$ with diagonal entries two and entries minus one along the tree. Number the trivalent node by 0. Its three arms have vertices (1), $(2,3)$, and $(4,5,6)$ for $E_7$, with $(4,5,6,7)$ as the third arm for $E_8$. All other entries are zero. Put $d = \det C$, so $d = 2$ or 1, respectively, and $B = dC^{-1}$. These are integral matrices with strictly positive entries.

Represent $d\omega_k$ in the simple-root basis by column $k$ of $B$. A simple reflection changes an integer vector $z$ by

$$z \longmapsto z - \left( \sum_l C_{il} z_l \right) e_i. \tag{4}$$

Moreover $\langle \omega_i, z \rangle = z_i$. Therefore the excluded pattern (3) becomes the exact pair of integer equalities

$$z_i = B_{ik}, \qquad z_j = -B_{jk}, \qquad i, j, k \text{ distinct.}$$

Starting at column $k$, close a set under all the reflections (4). Every stored vector belongs to the orbit, by induction on its construction. When no new vector is found, the set is invariant under every generator and contains the starting vector, so it is the full orbit. Thus no group-order estimate or randomized sampling enters the coverage argument.

The complete orbit certificates have the following sizes and contain no excluded pattern. A separate reader checks that each set contains its initial weight and is closed under every simple reflection. These two conditions already suffice for coverage of the required orbit.

| | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
|---|---|---|---|---|---|---|---|---|
| $E_7$ | 10080 | 576 | 2016 | 126 | 4032 | 756 | 56 | – |
| $E_8$ | 483840 | 17280 | 69120 | 2160 | 241920 | 60480 | 6720 | 240 |

These fifteen orbits contain 899402 vectors in total. The reader derives $B$ again by rational elimination from the displayed Dynkin trees. It checks strict ordering, uniqueness, the initial weight, every reflection image, and the absence of the excluded coordinate pattern. It does not import the orbit construction. The resulting finite sets are supplied in full, with deterministic compressed copies. The calculation enumerates the fundamental-weight orbits, rather than the groups of orders 2903040 and 696729600.

All arithmetic is integral. Indeed $z^T C z = dB_{kk}$ for an orbit vector; Cauchy–Schwarz gives $z_i^2 \le B_{ii} B_{kk} \le 900$. Every stored coordinate therefore has absolute value at most thirty. The producer checks this bound after each reflection; its intermediate integer expressions are bounded by the displayed matrices. A deadline or storage failure is an incomplete calculation and cannot return a complete certificate. The reader also rejects a set with a deleted vector, a changed coordinate, or an incorrect orbit header; the producer rejects a changed Cartan matrix. These controls exercise coverage and exact input identity.

## 6 Classification and proper residues

For clarity, the central antipode can be decided independently of the classification under discussion. In type $A_n$ with $n \ge 2$, a coordinate permutation on the sum-zero space cannot negate every difference $e_i - e_j$: three distinct coordinates already give contradictory required images. In $B_n$ every signed permutation occurs, whereas in $D_n$ only an even number of sign changes is allowed. Thus $-1$ occurs in $B_n$ and exactly the even $D_n$.

For $E_6$, the Cartan determinant is three. The weight lattice modulo the root lattice consequently has order three. Each simple reflection acts trivially on this quotient, since its difference from the identity is an integral multiple of a simple root. Negation acts nontrivially

on a group of order three, so it cannot belong to the Weyl group. The determinant and the usual weight/root-lattice identification are retained elementary Coxeter facts here.

For the remaining types, the accompanying exact matrix calculation verifies that the product of simple reflections, raised to the indicated power, is $-I$:

| type | $E_7$ | $E_8$ | $F_4$ | $H_3$ | $H_4$ |
|---|---|---|---|---|---|
| power | 9 | 15 | 6 | 5 | 15. |

The $E$ matrices are those already displayed. For $F_4$ use the chain Cartan matrix with diagonal two, neighbouring entries minus one except $C_{12} = -2$ (indices start at zero). For $H_3, H_4$ use the symmetric matrix with diagonal two, first bond $-\phi$, and other chain bonds minus one, where $\phi^2 = \phi + 1$. Reflection $i$ subtracts row $i$ of this matrix from row $i$ of the identity. Multiplication is exact in $\mathbb{Z}[\phi]$, or in $\mathbb{Z}$ in the integral cases. Faithfulness of these retained geometric representations identifies the computed matrix with the antipode in the Coxeter group. In rank one the antipodal quotient loses its generator. In rank two, quotienting $I_2(p)$ by a nontrivial central antipode reduces the order of the product of its generators, and hence changes the prescribed diagram.

The complete finite checks, Theorem 2, the string reconstruction, the even $D_n$ flag, and Lemma 4 give the following classification. The column $a$ records whether the antipodal quotient is also admitted. The group orders in the universal column are retained from the finite Coxeter classification, with the following correction to the fixed source. Its Table 1 prints $12 \cdot 6!$ for $E_6$, whereas the order is $51840 = 72 \cdot 6!$. This correction can be checked within the same orbit method. Number the $E_6$ tree by a central vertex 0 and arms (1), (2, 3), (4, 5). The certificate lists all twenty-seven vectors in the orbit of $3\omega_3$ and checks closure under the six reflections. The stabilizer is the standard parabolic obtained by deleting vertex 3, which has type $D_5$ and order $2^4 5! = 1920$. Orbit–stabilizer therefore gives $27 \cdot 1920 = 51840$. The same rational elimination verifies the Cartan determinant three used above.

| diagram | restriction | $\lvert W\rvert$ | $a$ |
|---|---|---|---|
| $A_n$ | $n \geq 1$ | $(n+1)!$ | 0 |
| $D_n$ | $n \geq 4$ | $2^{n-1}n!$ | 0 |
| $B_n$ | $n \geq 3$ | $2^n n!$ | 1 |
| $I_2(p)$ | $p \geq 3$ | $2p$ | 0 |
| $E_6$ | | 51840 | 0 |
| $E_7$ | | 2903040 | 1 |
| $E_8$ | | 696729600 | 1 |
| $F_4$ | | 1152 | 1 |
| $H_3$ | | 120 | 1 |
| $H_4$ | | 14400 | 1 |

There are exactly $1+a$ hypertopes in each row. When $a = 1$, the second object has distinguished group of order $|W|/2$. The two objects are distinct because they have different numbers of chambers.

Every proper residue of an admitted antipodal quotient is universal spherical. Indeed a maximal parabolic embeds, as proved above. Flag transitivity and the C-group intersection identities identify the residue of its corresponding face with the coset geometry of that embedded parabolic on its own maximal parabolics. This is its original spherical Coxeter complex. Residues of higher codimension are residues within these. Thus proper spherical residues are a consequence of the classification, not an assumption silently used to force the quotient kernel.

**Attribution and limits.** The classification conclusions and the standard Coxeter structure belong to the earlier literature. Maxwell's normal-subgroup theorem is a substantial retained input. The work here is the common marked rigidity argument, the explicit antipodal flag

criterion, the uniform $D_n$ obstruction, and the exceptional orbit proof with its exact finite obligations. No historical priority or open-conjecture solution is claimed. The Euclidean and hyperbolic classifications in the original manuscript lie outside the scope of this note.

## References


[1] M. E. Fernandes, D. Leemans, and A. Ivić Weiss, *An exploration of locally spherical regular hypertopes*. Fixed sixteen-page revised manuscript, catalogue L033; Sections 2–4.

[2] G. Maxwell, *The normal subgroups of finite and affine Coxeter groups*, Proc. London Math. Soc. 76 (1998), 359–382. Theorem 0.1 retained through the fixed source; primary full text not acquired.

[3] Human Compression project, *Residue stabilizers and distinguished generators*; *Kernels and intersections in quotients of hypertopes*. Complete reviewed components acquired with their proofs and stated inputs.

# Translation kernels in affine reflection groups

Human Compression project

**Abstract**

A normal subgroup of an irreducible affine Coxeter group which meets every maximal standard parabolic trivially consists of translations. The proof uses the commutator of an isometry with a wall reflection: nonparallel mirrors would give a forbidden finite-order element. This supplies Corollary 5.1 of the fixed hypertope manuscript under its Section 5 restriction to universal spherical residues. We separate that hypothesis from the weaker assertion that the residue diagrams are finite.

## 1 The geometric input

Let $W = \langle s_i : i \in I \rangle$ be an irreducible affine Coxeter group in its faithful action by Euclidean isometries on $E = \mathbb{R}^d$, where $d \geq 1$ and $|I| = d + 1$. Write $M_i$ for the mirror of $s_i$ and $W_i = \langle s_j : j \neq i \rangle$. We retain the usual affine Coxeter realization: $W$ is discrete, its closed fundamental alcove is a bounded simplex, every point is carried into that alcove by $W$, and the stabilizer of a point in the alcove is generated by the reflections in its containing walls. These are standard geometric Coxeter facts, not a classification of its normal subgroups. The linear normals to the walls span $E$; otherwise the alcove would contain a line. Their nonorthogonality graph is connected, as is the irreducible affine Coxeter diagram.

The translation subgroup is

$$T = \{w \in W : w(x) = x + b \text{ for some } b \in E\}.$$

No finite-index assumption will be placed on a subgroup of $W$.

**Lemma 1.** *If $N \triangleleft W$ and $N \cap W_i = 1$ for every $i \in I$, then $N$ is torsion-free.*

*Proof.* Suppose $g \in N$ has finite order $m$. For any $x \in E$, the average $m^{-1} \sum_{k=0}^{m-1} g^k x$ is fixed by $g$, since isometries are affine maps. Carry this point into the closed fundamental alcove. The conjugate of $g$ lies in $N$ by normality and in the stabilizer of that point. The containing walls are a proper subset of all alcove walls: a nondegenerate simplex has no point on every facet. Their generated group is therefore contained in some $W_i$. The assumed intersection forces the conjugate, and hence $g$, to be the identity. □

We also record why discreteness pays the other direction used below. The stabilizer of a point in a discrete isometry group is finite: after moving the point to the origin it is a discrete subgroup of the compact orthogonal group. If it were infinite, a convergent subsequence would give distinct group elements whose quotients tend to the identity, contradicting discreteness. Thus every element of $W$ fixing a point has finite order.

## 2 A reflection commutator

**Theorem 2.** *Let $W$ be as above. Every torsion-free normal subgroup of $W$ is contained in $T$. In particular,*

$$N \triangleleft W, \qquad N \cap W_i = 1 \ (i \in I) \quad \Longrightarrow \quad N \leq T.$$

*Proof.* Let $g \in N$ and write $g(x) = Ax + b$ with $A$ orthogonal. For each $i$, normality gives

$$c_i = g s_i g^{-1} s_i \in N.$$

Indeed $s_i g^{-1} s_i \in N$. The conjugate $g s_i g^{-1}$ is the reflection in $gM_i$.

Assume first $d \geq 2$. If $gM_i$ and $M_i$ are not parallel, they intersect. The product of their reflections fixes every point of that intersection. It therefore has finite order in the discrete group $W$. The mirrors are distinct, so their reflections are distinct and their product is not the identity. This contradicts torsion-freeness of $N$. Consequently $gM_i$ is parallel to $M_i$ for every $i$.

Choose a nonzero normal $u_i$ to each $M_i$. Parallelism and orthogonality of $A$ imply $Au_i = \varepsilon_i u_i$ with $\varepsilon_i \in \{1, -1\}$. If $\langle u_i, u_j \rangle \neq 0$, then

$$\langle u_i, u_j \rangle = \langle Au_i, Au_j \rangle = \varepsilon_i \varepsilon_j \langle u_i, u_j \rangle,$$

so $\varepsilon_i = \varepsilon_j$. Connectivity makes all signs equal. Since the normals span $E$, this proves $A = I$ or $A = -I$. In dimension one these are already the only orthogonal linear maps, so the same conclusion holds without the nonparallel-mirror argument.

If $A = -I$, then $g(x) = -x + b$ is a nonidentity involution, fixing $b/2$. This is again impossible in $N$. Hence $A = I$, and $g$ is a translation. The final implication follows from Lemma 1. □

The theorem uses normality twice, and irreducibility exactly in passing from preservation of the wall-normal lines to a common sign. It does not assert the conclusion for a general torsion-free subgroup, or for a reducible Euclidean reflection group.

## 3 The hypertope application and its precise domain

Let $\mathcal{H}$ be a regular hypertope with the irreducible affine Coxeter diagram of $W$. Its distinguished chamber involutions give a marked epimorphism

$$\pi : W \longrightarrow G = \mathrm{Aut}_I(\mathcal{H}).$$

Here we use the acquired residue-stabilizer proof: the involutions generate $G$, have the exact diagram labels, and each maximal marked parabolic $G_i$ acts faithfully and regularly on the corresponding residue. These conclusions do not assume that $\pi$ is locally injective.

**Corollary 3.** *Suppose every maximal residue of $\mathcal{H}$ is the universal spherical Coxeter complex with its inherited diagram. Then $N = \ker \pi$ lies in $T$, and $\mathcal{H}$ is the incidence quotient of the universal affine Coxeter complex by $N$.*

*Proof.* The restriction $W_i \to G_i$ is onto. The universal spherical residue hypothesis identifies its finite domain and image orders: both are the number of chambers of that spherical complex. Hence the restriction is injective, so $N \cap W_i = 1$. Theorem 2 gives $N \leq T$.

It remains to identify the incidence quotient, rather than only its group. An object of type $i$ in the universal complex is a left coset $wW_i$. Its $N$-orbit corresponds exactly to the coset $\pi(w)G_i$. Two quotient objects are incident when some representatives of their orbits are incident in the universal complex. This is equivalent to the two corresponding cosets in $G$ having a common representative: a common image lifts to $W$, and the discrepancy is absorbed by $N$. Thus the incidence quotient is the coset incidence system of $G$ with the subgroups $G_i$. The acquired reconstruction theorem for the actual regular hypertope identifies it with $\mathcal{H}$. □

This is a necessary-kernel statement for an already existing hypertope. An arbitrary normal translation subgroup need not give a hypertope. In particular, translations automatically avoid the finite parabolics, but can meet their products. The exact product-kernel condition and flag transitivity still have to be checked for a proposed quotient, as shown in the separately acquired correction to source Theorem 3.2. No converse existence or full period-lattice classification is asserted.

There is a genuine terminology issue in the fixed source. On page 5, "spherical" is defined by the finite Coxeter diagram, which also allows projective quotients. On page 10, Section 5 explicitly distinguishes spherical residues from projective residues and restricts attention to the former. Corollary 3 proves source Corollary 5.1 in this latter, locally universal sense. Finite residue diagrams alone do not imply the injectivity $W_i \to G_i$ used above; a projective residue has only half as many chambers. The source itself mentions a Euclidean example with projective residues on page 9. We neither silently exclude those examples from the earlier definition nor claim their kernels have been classified by the present argument. The same local hypothesis must be supplied when using the injectivity assertion in source Theorem 3.4.

**Attribution and verification.** The translation-kernel conclusion is the source's result, attributed there to Maxwell's affine normal-subgroup theorem. We have not acquired the full primary proof of that theorem and do not use it here. The new argument is the short reflection-commutator proof above, together with the exact residue interface. The affine realization and alcove-stabilizer facts remain named standard inputs. The supplementary exact matrix checks exercise the commutator step in two plane reflection groups; the uniform theorem is proved by the written argument, not by those finite examples. No historical priority, open-conjecture, new Lean or measurement claim is made.

## References


[1] M. E. Fernandes, D. Leemans and A. Ivić Weiss, *An exploration of locally spherical regular hypertopes*. Fixed sixteen-page revised manuscript, catalogue L033, pages 5, 7, 9–11.

[2] G. Maxwell, *The normal subgroups of finite and affine Coxeter groups*, Proc. London Math. Soc. 76 (1998), 359–382. The source attributes its affine conclusion to Theorem 0.2; this theorem is not a premise of the proof here.

[3] Human Compression project, *Residue stabilizers and distinguished generators*; *Kernels and intersections in quotients of hypertopes*. Complete acquired proofs with their distinct reviews and precise inputs.

# The rank-four toroidal families in cubic coordinates

Human Compression project

The fixed manuscript [1] displays three period families for each of the types $\widetilde{B}_3$ and $\widetilde{A}_3$ on page 11. They follow directly from the acquired cubic halving construction. We give the exact change of parameters, including the smallest allowed periods. This proves existence and the marked group descriptions of the displayed families. Their exhaustion is the classification attributed in the source to Ens; it is not a consequence of this parameter calculation.

## 1 The acquired construction

Put

$$D = \{v \in \mathbb{Z}^3 : v_1 + v_2 + v_3 \equiv 0 \pmod 2\}, \qquad E = 2\mathbb{Z}^3 + \mathbb{Z}\mathbf{1}.$$

Write $B_3$ for the signed permutation group, and $B_3^+$ for the subgroup with an even number of sign changes. Let $z_i$ change the sign of coordinate $i$, let $s_i$ interchange coordinates $i, i+1$, and let $\tau_v$ be translation by $v$. The acquired affine calculation gives

$$x = \tau_{(1,1,0)} z_1 z_2 s_1, \qquad y = z_2 z_3 s_2,$$

$$\langle x, s_1, s_2, z_3 \rangle = D \rtimes B_3, \qquad \langle x, s_1, s_2, y \rangle = D \rtimes B_3^+.$$

Their diagrams have, respectively, the bonds $02, 12$ labelled three and $23$ labelled four, and the four-cycle $02, 12, 13, 03$ with all labels three. Thus these are the marked affine groups of types $\widetilde{B}_3$ and $\widetilde{A}_3$.

For $t \geq 2$ put

$$\Lambda_1(t) = t\mathbb{Z}^3, \qquad \Lambda_2(t) = tD, \qquad \Lambda_3(t) = tE, \qquad L_k(t) = D \cap \Lambda_k(t).$$

The acquired geometry proof applies one halving at the left end of the cubic toroid, and then one at the opposite end. It gives actual regular hypertopes $H_{10}$ and $H_{11}$, with the above markings and groups

$$(D/L_k(t)) \rtimes B_3, \qquad (D/L_k(t)) \rtimes B_3^+,$$

for every $k \in \{1, 2, 3\}$ and $t \geq 2$ except $(k, t) = (1, 2)$. Here existence includes thinness, residual connectivity and chamber transitivity, proved in the acquired incidence models. These conclusions are not being inferred from the diagrams or group orders. In particular, the case $(k, t) = (2, 2)$ is allowed for $H_{10}$ and $H_{11}$: its separate exclusion concerns a second halving at the same end.

## 2 The source parameters

The period generated by a source vector means the integer span of its orbit under the corresponding finite linear group. Both finite groups above give the same three spans in dimension three.

**Proposition 1.** *The following table identifies every displayed source family with the acquired construction. The orders in the last two columns are the orders of the full type-preserving groups.*

| *source vector and domain* | $(k,t)$ | $L_k(t)$ | $\lvert G_{\widetilde{B}_3}\rvert$ | $\lvert G_{\widetilde{A}_3}\rvert$ |
|---|---|---|---|---|
| $(2s,0,0),\ s\geq 2$ | $(1,2s)$ | $2s\mathbb{Z}^3$ | $192s^3$ | $96s^3$ |
| $(s,s,0),\ s\geq 2$ | $(2,s)$ | $sD$ | $48s^3$ | $24s^3$ |
| $(2s,2s,2s),\ s\geq 1$ | $(3,2s)$ | $2sE$ | $768s^3$ | $384s^3$ |

*The source objects are the corresponding incidence quotients of the universal marked affine complexes.*

*Proof.* The orbit of $e_1$ under either group is the set of signed coordinate vectors: an unwanted sign change can be placed in a zero coordinate. Its span is $\mathbb{Z}^3$. Similarly the orbit of $e_1+e_2$ consists of all roots $\pm e_i \pm e_j$. These span $D$: for example $e_1+e_2, e_2-e_1, e_3-e_2$ are a lattice basis of determinant two.

Under $B_3$, the orbit of $\mathbf{1}$ spans $E$, since sign differences give $2e_i$. Under $B_3^+$ it spans $M = 2D+\mathbb{Z}\mathbf{1}$: differences from $\mathbf{1}$ are twice the characteristic vectors of two-element subsets, whose span is $D$. In dimension three $M=E$. Indeed $\mathbf{1}-e_i \in D$, and $2e_i = 2\mathbf{1}-2(\mathbf{1}-e_i) \in M$. The reverse inclusion is immediate. Scaling gives precisely the three source period lattices in the table.

Each of these lattices lies in $D$. The first and third do so because their coordinates are even, and the middle does by definition. Consequently their intersections with $D$ are unchanged. Substitution in $L_k(t)$ proves the parameter correspondence. The first row has $t\geq 4$, the second has $t\geq 2$ with $k=2$, and the third has $t\geq 2$ with $k=3$. None is the excluded pair $(1,2)$, so both acquired geometry constructions apply throughout the exact source domains, including $s=1$ in the third row.

Finally $[\mathbb{Z}^3 : D] = 2$, $[\mathbb{Z}^3 : E] = 4$, and scaling a lattice by $a$ multiplies its index by $a^3$. Thus the three indices in $D$ are $4s^3, s^3, 16s^3$. Multiply these by $|B_3| = 48$ or $|B_3^+| = 24$. The acquired coset identification of each actual geometry gives the stated incidence quotient, with its actual marking. □

**Attribution and scope.** The source attributes the complete rank-four family classification to Ens. The cubic toroid input is the retained McMullen–Schulte classification, and the halving constructions originate with the earlier authors cited in the acquired proof. The present addition is the literal parameter and marking interface between those proofs and this source. It does not prove that every invariant lattice is in the table, relax a small-period condition, or claim a new conjecture answer. No new finite census, Lean formalization or compression measurement is involved.

# References


[1] M. E. Fernandes, D. Leemans and A. Ivić Weiss, *An exploration of locally spherical regular hypertopes*, fixed catalogue L033 manuscript, pages 10–11.

[2] Human Compression project, *Generalized halving: corrected source recovery*, accepted affine group calculation and iterated geometry proof, Sections 20 and 30 in the frozen application sources.

# Finite quotients preserving spherical residues

Human Compression project

**Abstract**

For a Coxeter system with finite maximal parabolics, a finite set of products of three parabolics detects every failure of flag lifting in a quotient. Avoiding this set preserves the entire incidence structure of each proper residue, as well as the intersection property. Reduction of the faithful geometric representation modulo a suitable prime produces such quotients of arbitrarily large order whenever the Coxeter group is infinite. This gives a common proof of the finite existence assertions in Section 6 of the fixed manuscript, including the diagrams which are not strings. It does not determine the smallest quotient or verify the groups in the source's finite table.

## 1 A finite obstruction set

Let $(W, S)$ be a Coxeter system of rank $r \geq 3$, with $S = \{s_i : i \in I\}$. Put $W_J = \langle s_j : j \in J \rangle$ for $J \subseteq I$, and write $P_i = W_{I\setminus\{i\}}$. Assume that every $P_i$ is finite. We use the standard Coxeter complex: its objects of type $i$ are the left cosets $wP_i$, and two objects of different types are incident when their cosets intersect. We retain the classical Coxeter facts that $W_J \cap W_K = W_{J\cap K}$, that this incidence system is a thin, residually connected geometry, and that $W$ acts regularly on its chambers. In particular every pairwise incident set extends to a chamber. These are inputs about the universal complex, not assumptions about a proposed quotient.

Define the finite set

$$\mathcal{F} = \left( \bigcup_{i,j,k\in I} P_i P_j P_k \right) \setminus \{1\}.$$

Repetitions among $i, j, k$ are allowed. Since every subgroup contains the identity, this set also contains all nonidentity elements in any $P_i$ or $P_i P_j$.

**Theorem 1.** *Let $N \triangleleft W$ satisfy $N \cap \mathcal{F} = \varnothing$, and let $q : W \to Q = W/N$. The marked group $(Q, (q(s_i))_{i\in I})$ has the full intersection property. Its coset incidence system with maximal subgroups $q(P_i)$ is a regular hypertope. The residue of its base object of type $i$ is isomorphic, as a typed incidence geometry, to the residue of $P_i$ in the universal Coxeter complex. Thus every proper residue remains universal and spherical. The quotient map preserves every bond label.*

*Proof.* First, $q$ is injective on every $P_i$. More generally, let $J, K$ be proper subsets of $I$ and choose $i \notin J$, $j \notin K$. If $a \in W_J$, $b \in W_K$ and $q(a) = q(b)$, then $ab^{-1} \in N \cap P_i P_j$, so $a = b$. The intersection property in $W$ therefore gives

$$q(W_J) \cap q(W_K) = q(W_{J\cap K}).$$

When either subset is all of $I$, this equality is immediate. This proves the full intersection property, including the trivial intersection of all maximal marked parabolics. Every subgroup generated by two marks is contained in a maximal parabolic, since $r \geq 3$. Its injection proves that no mark disappears and that all pairwise product orders are unchanged.

We next prove the required flag lifting, rather than inferring it from these subgroup equalities. Let a nonempty flag in the quotient contain an object of type $i$. Translate it so that this object is $q(P_i)$. Every other object, of type $j$, is incident with $q(P_i)$, and hence has a representative

$$q(a_j)q(P_j), \qquad a_j \in P_i.$$

Choose such a representative separately for every type in the flag. The corresponding objects $a_jP_j$ are all incident with $P_i$. For distinct types $j, k$ in the flag, quotient incidence says

$$q(a_j^{-1}a_k) \in q(P_jP_k).$$

Choose $b \in P_j$, $c \in P_k$ with $q(a_j^{-1}a_k) = q(bc)$. Then

$$a_j^{-1}a_kc^{-1}b^{-1} \in N \cap P_iP_kP_j.$$

The avoidance hypothesis makes this element the identity. Consequently $a_j^{-1}a_k = bc \in P_jP_k$, which is exactly the incidence of $a_jP_j$ and $a_kP_k$. Thus the chosen objects form an actual flag upstairs. They extend to a chamber of the Coxeter complex. Its image is a chamber containing the prescribed quotient flag. The same argument proves that every quotient flag is a translate of a subflag of the base chamber: lift, extend and use the chamber action of $W$. In particular $Q$ is flag transitive.

Here is the asserted residue isomorphism in detail. An object of type $j \neq i$ incident with $P_i$ has the form $aP_j$ for $a \in P_i$. Mapping it to $q(a)q(P_j)$ is surjective onto the corresponding quotient residue. If $a, b \in P_i$ give the same image object, then $q(a^{-1}b) \in q(P_j)$. The already proved intersection equality and injectivity on $P_i$ imply $a^{-1}b \in P_i \cap P_j$, so the objects upstairs coincide. This proves injectivity on objects. The preceding triple-product calculation proves that incidence of any two image objects reflects incidence upstairs. Incidence is preserved automatically. Hence the map is an isomorphism of the complete residue geometries, not merely of their groups or their chamber sets.

Every nonempty proper residue in the quotient is therefore a residue of a finite spherical Coxeter complex. It is thin and, when its rank is at least two, connected. The whole quotient geometry is connected as well: its chambers are the translates of the base chamber, with trivial chamber stabilizer by the intersection property. Multiplying successively by the marks gives galleries joining these chambers, because the marks generate $Q$. A gallery gives connectedness of the incidence graph in rank at least two. Thinness and residual connectedness now hold in all required ranks. Finally, in this geometry, a type-preserving automorphism fixing a chamber fixes its adjacent chambers and then every chamber, by the gallery connectedness just proved. Thus the full type-preserving automorphism group acts freely on chambers. Since $Q$ already acts transitively and freely, it is that full group. □

The third factor has a specific role: two objects represented inside one fixed parabolic may become incident after quotienting. Pairwise parabolic injection alone does not rule this out. The argument does not claim that the displayed avoidance condition is necessary, or that it is economical for finding a small quotient.

## 2 Separating finitely many matrices

We give the elementary finite-quotient argument needed here. It also specifies what is retained from Coxeter theory.

**Lemma 2.** *Let $K$ be a number field with ring of integers $\mathcal{O}_K$, and let $G \leq \mathrm{GL}_d(\mathcal{O}_K)$. For every finite set $A \subset G \setminus \{1\}$ there is a homomorphism from $G$ to a finite group whose kernel is disjoint from $A$.*

*Proof.* For every $g \in A$, choose a nonzero entry $a_g$ of $g - I$. Its algebraic norm $\mathrm{Norm}_{K/\mathbb{Q}}(a_g)$ is a nonzero integer: it is the determinant of multiplication by $a_g$ on any integral basis of $\mathcal{O}_K$. Choose a rational prime $p$ not dividing the product of these finitely many integers; for empty $A$ any prime will do. Modulo $p$, each multiplication matrix remains invertible. In particular $a_g$ has nonzero image in the finite ring $\mathcal{O}_K/p\mathcal{O}_K$.

Reduction of matrices is a homomorphism

$$G \longrightarrow \mathrm{GL}_d(\mathcal{O}_K/p\mathcal{O}_K).$$

Inverses remain inverses after reduction. The target is finite, since the additive group of the coefficient ring has $p^{[K:\mathbb{Q}]}$ elements. The chosen entry of every $g - I$ survives, so none of the elements of $A$ maps to the identity. □

For a finite-rank Coxeter system the standard geometric representation has, in its simple-root basis, the matrices

$$s_i(e_i) = -e_i, \qquad s_i(e_j) = e_j + 2\cos(\pi/m_{ij})e_i \quad (j \neq i).$$

The usual convention for $m_{ij} = \infty$ replaces the coefficient by 2. We retain the faithfulness theorem for this representation. Each finite-label coefficient is an algebraic integer: $2\cos(\pi/m) = \zeta_{2m} + \zeta_{2m}^{-1}$. There are only finitely many labels, so all coefficients lie in the ring of integers of a single cyclotomic number field. The generators are involutions and have determinant $-1$. Thus both the matrices and their inverses have integral coefficients, and Lemma 2 applies. This proves the particular residual-finiteness assertion used below without invoking a normal-subgroup classification.

**Corollary 3.** *Let $(W, S)$ be an infinite Coxeter system of finite rank at least three, with every maximal standard parabolic finite. For every positive integer $M$ there is a finite regular hypertope with this diagram, with universal spherical proper residues, and with at least $M$ chambers.*

*Proof.* Choose $M$ distinct elements $w_1, \ldots, w_M$ of $W$ and enlarge the finite set $\mathcal{F}$ by the nonidentity elements $w_a^{-1}w_b$ for $a \neq b$. Apply the faithful representation and Lemma 2 to separate this entire set simultaneously. The image group $Q$ is finite, and its kernel satisfies Theorem 1. The images of the $w_a$ are distinct, so $|Q| \geq M$. The regular chamber action identifies $|Q|$ with the number of chambers of the resulting hypertope. □

This proof is effective in an elementary sense. The finite parabolics can be enumerated in the exact representation, their triple products formed, and a nonzero matrix entry selected for every forbidden element. The norm calculation then supplies a prime to avoid. The finite matrix image can itself be enumerated. These steps need not be computationally practical; no small-order bound is asserted. Taking unbounded $M$ gives infinitely many pairwise nonisomorphic finite hypertopes, rather than just many maps onto the same object.

## 3 Source consequences and attribution

Section 6 of the fixed manuscript [1] recalls infinitely many finite maps and hypermaps for the compact hyperbolic triangle diagrams. It also recalls, from the classical polytope literature, infinitely many finite locally spherical polytopes for the string diagrams in ranks four and five. Corollary 3 supplies one proof for these assertions. Indeed a compact hyperbolic Coxeter simplex has finite proper parabolics, and its reflection group is infinite. The same argument applies to the nonstring compact hyperbolic diagrams in the source's Table 2. We retain that diagram classification; the corollary does not provide a new proof that the list

is exhaustive. For a string diagram the acquired string coset correspondence identifies the resulting hypertope with a regular abstract polytope.

The conclusion uses actual universal spherical residues. The source also uses "spherical" for the weaker finite-diagram condition, which may permit projective residues. Our existence result satisfies the stronger condition. It cannot be used to rule out smaller examples with projective residues. Nor does it identify a quotient group with any particular extension listed in Table 3, prove a minimum there, or verify a quoted exceptional construction. Those remain separate finite or attributed inputs.

The finite existence results for string diagrams are prior work, as are residual finiteness of Coxeter groups and the standard Coxeter complex. The contribution of this component is the explicit triple-product argument linking a finite separating set to full flag lifting and residue incidence, followed by the common application. No claim of historical priority is made for this argument. The earlier project note *Finite quotients with prescribed parabolic intersections* treats different hypotheses involving separability and endomorphisms; none of its unproved filling assertions is used here. This component makes no claim to resolve a source conjecture, complete the entire L033 manuscript, provide a Lean formalization, or establish a numerical compression factor.

## References


[1] M. E. Fernandes, D. Leemans and A. Ivić Weiss, *An exploration of locally spherical regular hypertopes*, fixed sixteen-page revised manuscript, catalogue L033, Section 6 and Table 2. Its classical existence citations and the finite table are distinguished above.

[2] Human Compression project, *Residue stabilizers and distinguished generators*; *Kernels and intersections in quotients of hypertopes*. The present proof retains the universal Coxeter complex and its standard intersection property, and proves the quotient interfaces explicitly.

# Eight bounded rank-four hyperbolic quotient searches

Independent recovery of fixed-source Table 3

**Abstract**

We certify finite marked quotients for eight diagrams appearing in Table 3 of Fernandes–Leemans–Weiss. Complete bounded normal-subgroup searches, followed by intersection and actual coset-incidence tests, give 59 positive marked kernels. We distinguish universal spherical residues from residues whose diagrams are finite. In particular no regular hypertope with either local convention realizes the printed order 336 for $(3,4,3,4)$ or order 2160 for $(3,5,3,4)$. Exact named extension descriptions and the ninth diagram $(3,5,3,5)$ are outside this component.

## 1 Conventions and the finite assertion

Write $I = \{1,2,3,4\}$. For a symmetric Coxeter matrix $M$, put

$$W(M) = \langle s_i \ (i \in I) \mid s_i^2 = 1, (s_i s_j)^{M_{ij}} = 1 \ (i < j) \rangle.$$

In a square $(p,q,r,s)$ the edges $12, 23, 34, 41$ have those labels, and the diagonals have label 2. In a string $[p,q,r]$ the edges are $12, 23, 34$. For the fork $[5,3_3]$ the edges $12, 13, 14$ have labels $5, 3, 3$. Unlisted off-diagonal entries are 2. These conventions agree, up to relabelling, with the native diagrams on source physical page 12 and the types on physical page 14.

For a quotient $H = \langle t_1, t_2, t_3, t_4 \rangle$, let $H_J = \langle t_j : j \in J \rangle$ and $P_i = H_{I \setminus \{i\}}$. The *broad* column below requires a regular hypertope with this exact diagram; its rank-three residues have finite diagrams. The *universal* column additionally requires each $P_i$ to be the full finite Coxeter group for the corresponding rank-three subdiagram. This distinction matters: the fixed source defines spherical by a finite diagram on page 5, but explicitly restricts to spherical, rather than projective, local geometries in its euclidean discussion on page 10. We do not silently resolve that terminological tension for Section 6.

| Diagram | Bound | Kernels | Positive | Broad minimum | Universal minimum |
|---|---|---|---|---|---|
| $(3,3,3,4)$ | 7680 | 14 | 10 | 168 | 1920 |
| $(3,3,3,5)$ | 7200 | 5 | 2 | 2520 | 7200 |
| $(3,4,3,4)$ | 9216 | 45 | 21 | 168 | 1152 |
| $(3,5,3,4)$ | 2160 | 8 | 1 | 1080 | $> 2160$ |
| $[3,5,3]$ | 8160 | 5 | 2 | 660 | 8160 |
| $[4,3,5]$ | 7680 | 17 | 7 | 1920 | 3840 |
| $[5,3,5]$ | 8160 | 6 | 3 | 3420 | 7200 |
| $[5,3_3]$ | 7680 | 20 | 13 | 960 | 3840 |

Here kernels include the whole group (index 1). Positive means distinct normal kernels for the fixed ordered presentation, not isomorphism classes after diagram automorphisms. There are 120 kernels in total and 59 positives. The entry $> 2160$ is only an exclusion through that bound, not an existence or minimum assertion beyond it. All displayed attained minima have explicit permutation witnesses. The original source's group names are not inferred from orders or from descriptive strings printed by GAP.

## 2 A complete coset test

Suppose the involutions $t_i$ generate a finite group $H$, have exact pair orders $M_{ij}$, and satisfy

$$H_A \cap H_B = H_{A\cap B} \qquad (A, B \subseteq I). \tag{1}$$

Form the incidence system whose objects of type $i$ are right cosets $P_i g$, with incidence given by nonempty intersection. For $J \subseteq I \setminus \{k\}$ let $E(J,k)$ be the set of type-$k$ objects incident with every base object $P_i$, $i \in J$. The following exact test is applied for all 32 choices:

$$E(J,k) = \{P_k h : h \in H_{I\setminus J}\}. \tag{2}$$

The GAP implementation compares the actual sets in the right-coset permutation action. The independent integer-permutation implementation constructs every right coset as a literal set and checks the equivalent cardinality $|E(J,k)| = |H_{I\setminus J}|/|H_{I\setminus(J\cup\{k\})}|$. The orbit on the right of (2) is always contained in the left, and its stabilizer is the denominator subgroup by (1); consequently this cardinality check really implies set equality.

We give the argument that this tests flag-transitivity, including flags which might otherwise be omitted by a chamber-only model. Induct on the size of a flag. Normalize its first $|J|$ objects to the base flag using the induction hypothesis. Any further incident type-$k$ object lies in $E(J,k)$ and hence can be normalized by the base-flag stabilizer $\bigcap_{i\in J} P_i = H_{I\setminus J}$. Thus every flag is an image of a base flag. In particular every flag extends to a chamber, and $H$ is flag-transitive on the actual incidence system.

The chamber stabilizer is trivial by (1). A corank-one flag has exactly two completions because its stabilizer is the two-element group generated by its remaining mark. More generally, the stabilizer of a base flag is generated by the remaining marks. Their chamber-adjacency graph is connected. For a residue of rank at least two these successive adjacent chambers share an object, so its incidence graph is connected. The system is therefore a thin, residually connected geometry. Finally any type-preserving automorphism fixing a chamber fixes its unique adjacent chamber of each type and then every chamber. It is the identity. Consequently the full type-preserving group is precisely $H$, and the geometry is a regular hypertope.

All maximal subdiagrams here are finite. Their universal orders, in our chosen vertex order, are respectively

$$\begin{array}{ll} 3334 : (24,48,48,24), & 3335 : (24,120,120,24), \\ 3434 : (48,48,48,48), & 3534 : (120,48,48,120), \\ 353 : (120,12,12,120), & 435 : (120,20,16,48), \\ 535 : (120,20,20,120), & 5333 : (8,24,120,120). \end{array}$$

These are the orders of $A_3, B_3, H_3$ or products of rank-one and dihedral Coxeter groups. Equality with these orders makes the natural surjection of each finite local Coxeter group injective. Otherwise the actual local orders are retained explicitly in the witness records; the broad column does not assume an injective local quotient.

## 3 Exhaustion, source corrections, and retained premises

Every finite regular hypertope with the prescribed diagram supplies its distinguished involutions, hence an epimorphism $W(M) \to H$. Its kernel is normal and has index $|H|$. This implication needs no injection on local parabolics. Relabelling a source-equivalent diagram gives a marking of our fixed presentation, so does not evade the normal-kernel search. We enumerate all normal subgroups

through the displayed bound, discard quotients with wrong mark or pair orders, and apply (1) and (2). The only index-only prefilter is divisibility by $\mathrm{lcm}_{i<j}(2M_{ij})$: each genuine rank-two marked subgroup has order $2M_{ij}$. Thus no required quotient is lost.

The enumeration uses GAP 4.15.1 and LINS 0.9, `LowIndexNormalSubgroupsSearchForAll`, with the unmodified default options. Its documented contract is all normal subgroups through the index bound, not one quotient per abstract isomorphism type. The installed limit is $10^7$, checked before each search. The default has no early termination or branch cut, and uses `UseLIS=false`. The algorithm and its exact computer implementation are retained computational premises. In particular LINS's tables of characteristically simple groups, Schur multiplier primes, and transitive socle-overgroups are classification inputs; we do not claim to reprove or regenerate their completeness. The local documentation attributes the algorithm to Firth's 2005 thesis and explains its pregenerated tables. Exact executable/table bytes and clean terminal receipts are bound in the packet. This is not a formal verification of GAP or LINS.

There are exactly three kernels of index 336 for the first disputed square and one of index 2160 for the second. Every resulting marking fails (1), even allowing smaller finite-diagram local residues. Their compact tuples are retained and independently checked by literal permutation closure. The source order 336 and 2160 rows therefore do not describe regular hypertopes of their printed diagrams. A further negative control is the order 360 quotient of $(3, 5, 3, 4)$: it satisfies all 256 intersections but has four, rather than two, type1 completions of a particular rank-three base flag. It illustrates why local orders or the intersection property alone cannot replace the incidence test.

There is also an elementary, more restricted obstruction at 336: with four universal $B_3$ residues, choose two maximal parabolics whose remaining edge has label3. Their intersection would have order 6, and their product set would have $48^2/6 = 384$ elements. This argument alone does *not* exclude projective local residues; the unqualified index 336 exclusion above uses all three kernels.

Fresh and retained positive inventories are compared by a bijection of ordered markings: an explicitly checked isomorphism sends each of the four old marks to the corresponding new mark. This is equivalent to equality of their kernels in the fixed presentation. Every record is matched once, so multiplicities are preserved; equality of compact permutation bytes is neither required nor presumed. The literal JSON inputs are also compared with the exact GAP records before their independent audit.

The positive checks are repeated using independent standard-library permutation closure and explicit coset sets. Neither compacting the GAP actions nor printing structural descriptions supplies a completeness premise. This component proves the finite assertions displayed above; it does not finish Table 3's exact named extensions, the ninth square, rank-five existence, the source's other classifications, or the whole original. Initial universal-only checks and all capped square 3535 attempts are preserved as history and are not substituted for the broad complete executions.

**Attribution.** The target rows, diagrams and minimum claims belong to Fernandes–Leemans–Weiss, *An Exploration of Locally Spherical Regular Hypertopes*, fixed 16-page source, Sections 2 and 6, Tables 2–3. The parent supplied the first local-order square 3434 probe and identified the source's local-scope ambiguity. The present author supplied the full incidence/intersection implementation, independent literal audit and bounded core synthesis. A separate reader inspected the LINS documentation and table boundary. None of these activities asserts priority for the underlying finite-group algorithms or adds a new measurement or question.

# Group certificates for twenty-two hyperbolic table entries

Finite companion to the fixed-source recovery

**Abstract**

We identify actual automorphism groups of twenty-two marked quotients already certified as regular rank-four hypertopes. Explicit homomorphisms, normal kernels and complements establish the group descriptions in Table 3 of Fernandes–Leemans–Weiss. The proof uses no new normal-subgroup search. The two excluded orders and the ninth diagram remain separate obligations of the complete source contract.

## 1 The precise assertion

Put $E_d = \mathrm{C}_2^d$, $L_q = \mathrm{PSL}_2(q)$, and let $X_{32}$ denote the actual extraspecial group of order32 certified below. A colon means a split extension, a cross a direct product, and a dot a nonsplit extension. An expression with an unspecified action denotes the actual extension supplied by its certificate; it is not a uniqueness assertion about every group with that order or series.

We acquire the complete eight-type quotient packet and its distinct review and root adoption. Its record $(\tau, j)$ is an ordered four-tuple of permutations; $H_{\tau,j}$ is the literal group generated by that tuple. The packet proves the exact Coxeter orders, all256 subset intersections and actual flag transitivity for every selected record. Thus these properties are not inferred here from the abstract group name. Its numbering is fixed; a freshly enumerated kernel list need not have the same order.

**Theorem 1.** *For each row of Table 1, the actual marked group $H_{\tau,j}$ has the displayed description. Consequently all twenty-two positive named entries of fixed-source Table 3 outside the diagram $(3,5,3,5)$ and the two excluded orders are realized by the previously certified hypertopes.*

The first four digits denote successive square edges; three digits denote a string. The label $5;3,3$ denotes the star with three edges at vertex1. The source's "locally spherical" diagram condition and the stronger universal-residue condition remain distinct, as in the acquired core. The source group notation need not uniquely fix an action. For these actual twenty-two records, the stored local orders equal the universal orders in the acquired core. Thus these witnesses also have universal rank-three residues; this follows from their exact local data, not their group names.

## 2 Certificate principles

For finite permutation groups $N, K \leq D$, the equations

$$N \lhd D, \qquad N \cap K = 1, \qquad \langle N, K \rangle = D$$

give $D = N \rtimes K$. A specified bijective homomorphism $K \to M$ identifies the complement with the named model $M$. An involution $t \notin D$ with $D \lhd \langle D, t \rangle$ and index2 gives another split step. If $z$ is a central involution outside the preceding subgroup $U$, then $\langle U, z \rangle = U \times \langle z \rangle$. These statements follow immediately from the unique product expression in subgroups with trivial intersection; they require actual containment and normality, which the checker tests explicitly.

| Diagram $\tau$ | Record $j$ | Order | Certified description |
|---|---|---|---|
| $(3,3,3,4)$ | 8 | 1920 | $E_4 : \mathrm{A}_5 : \mathrm{C}_2$ |
| | 9 | 2688 | $(E_3 : L_7) \times \mathrm{C}_2$ |
| | 11 | 3840 | $E_5 : \mathrm{A}_5 : \mathrm{C}_2$ |
| | 12 | 7680 | $(\mathrm{C}_4 \,.\, (E_4 : \mathrm{A}_5)) : \mathrm{C}_2$ |
| | 13 | 7680 | $E_6 : \mathrm{A}_5 : \mathrm{C}_2$ |
| $(3,3,3,5)$ | 5 | 7200 | $(\mathrm{A}_5 \times \mathrm{A}_5) : \mathrm{C}_2$ |
| $(3,4,3,4)$ | 24 | 1152 | $X_{32} : \mathrm{S}_3 : \mathrm{S}_3$ |
| | 27 | 1440 | $\mathrm{S}_6 \times \mathrm{C}_2$ |
| | 36 | 4320 | $\mathrm{S}_6 \times \mathrm{S}_3$ |
| | 38 | 5376 | $(E_3 : L_7) \times \mathrm{C}_2^2$ |
| | 40 | 6048 | $L_7 \times \mathrm{S}_3^2$ |
| | 45 | 9216 | $E_8 : \mathrm{S}_3 : \mathrm{S}_3$ |
| $[3,5,3]$ | 5 | 8160 | $L_{16} : \mathrm{C}_2$ |
| $[4,3,5]$ | 14 | 3840 | $E_5 : \mathrm{A}_5 : \mathrm{C}_2$ |
| | 12 | 3840 | $(E_4 : \mathrm{A}_5 : \mathrm{C}_2) \times \mathrm{C}_2$ |
| | 15 | 7680 | $(E_5 : \mathrm{A}_5 : \mathrm{C}_2) \times \mathrm{C}_2$ |
| | 16 | 7680 | $E_5 : \mathrm{A}_5 : \mathrm{C}_2 : \mathrm{C}_2$ |
| $[5,3,5]$ | 5 | 7200 | $(\mathrm{A}_5 \times \mathrm{A}_5) : \mathrm{C}_2$ |
| | 6 | 8160 | $L_{16} : \mathrm{C}_2$ |
| $[5;3,3]$ | 14 | 3840 | $(E_5 : \mathrm{A}_5) \times \mathrm{C}_2$ |
| | 19 | 3840 | $E_6 : \mathrm{A}_5$ |
| | 20 | 7680 | $(E_6 : \mathrm{A}_5) \times \mathrm{C}_2$ |

Table 1: Actual tuples in the immutable eight-type core.

For the three $E_d : \mathrm{A}_5 : \mathrm{C}_2$ square rows the checker in fact constructs an $\mathrm{S}_5$ complement. Its normal alternating subgroup and a transposition give the displayed successive split extensions. The two complements of order36 are explicitly $\mathrm{S}_3 \times \mathrm{S}_3$, so their displayed two-step series also follows, with the actual induced actions.

For $X_{32}$ the checker finds $|X_{32}| = 32$ and $Z(X_{32}) = X_{32}'$ of order2, with generator squares in this central subgroup. The quotient is elementary abelian: its generators commute and have order at most2. Thus $X_{32}^2 X_{32}' = Z(X_{32})$; the elementary Frattini formula for a finite2-group shows that the Frattini subgroup also equals this order2 group. This proves the asserted extraspecial property; no sign or small-group identifier absent from the source is inferred.

For the affine rows we construct translations of $\mathbb{F}_2^3$ and the literal natural $\mathrm{GL}_3(2)$ action on its eight vectors. A retained bijective generator homomorphism identifies the linear group with $L_7$. The resulting affine group has order1344. The required direct products are then named models, compared by explicit bijective homomorphisms to the entire actual $H$. No group is identified merely by its order.

## 3 The central nonsplit row and execution interface

For $(\tau, j) = ((3,3,3,4), 12)$ the certificate gives

$$D \triangleleft H, \quad |D| = 3840, \quad [H : D] = 2, \quad Z = \mathrm{C}_4 \leq Z(D), \quad Z \cap D' \neq 1.$$

It also gives $N \triangleleft D$ of order64 containing $Z$, and a subgroup $U \leq D$ with a surjective homomorphism $U \to \mathrm{A}_5$ whose kernel is exactly $Z$. Generator squares and commutators of $N$ lie in $Z$, so $N/Z \cong E_4$. The checked equalities $N \cap U = Z$ and $NU = D$ give $D/Z \cong E_4 : \mathrm{A}_5$. If the central extension by $Z$ split, then $D = Z \times K$ for a complement $K$, and $D' = K'$ would be disjoint from $Z$. This contradicts $Z \cap D' \neq 1$, proving the dot assertion. An actual involution outside $D$ generates $H$ with $D$, proving the final split factor. The other split $L_{16}$ rows additionally satisfy $C_H(D) = 1$ for their normal $D \cong L_{16}$; their external action is therefore faithful, rather than an unproved identification by order.

**Finite verification.** The three certificate files contain only literal permutations, specified subgroup generators, homomorphism images and outside elements. The verifier reads the original core tuple and reconstructs every group from generators. It checks containment, exact orders, normality, intersections, elementary abelian tests, each claimed full generated group, and actual bijectivity of each generator homomorphism. The twelve series records, nine direct-model records and one central-extension record have exactly the22 distinct keys in Table 1. Five damaged certificates independently corrupt a model image, an elementary kernel, an outside involution, the order4 central kernel, or the linear-group bridge; each is rejected.

These are direct positive certificates. Their earlier search routines are candidate providers only, and their execution is not a completeness premise. The accepted core separately supplies the necessary incidence mathematics and complete normal-quotient searches. Default replay checks all22 groups and the five damaged inputs; it does not rerun those searches. Python optimization is rejected before output. Complete native sources are included.

**Sources, software and scope.** The source is M. Fernandes, D. Leemans and A. I. Weiss, the fixed sixteen-page L033 article, Table 3 on physical page14. Its positive group entries are the objects recovered here; we make no priority claim. The exact permutation, homomorphism and finite-field calculations use GAP4.15.1. Its documented $\mathrm{PSL}(2, q)$ constructor is the action of determinant-one matrices on projective lines, and the standard alternating/symmetric constructors are named-model interfaces. These software interfaces are retained explicitly, not formally verified by this computation. The affine model itself is built from the stated vector action. The acquired LINS completeness and finite classification premises are unchanged and are not used for a new census here.

The project root supplied the original normal-quotient probe; the publication agent authored these group certificates, verifier and companion proof. Earlier distinct core reviews are acquired with their original roles. The two source orders336 and2160 are excluded by that core, not by this naming argument. The ninth diagram, its two printed orders3840 and9720, and all remaining whole-source obligations are outside this theorem.

# Restricting a normal quotient search to divisors of its target order

We use the previously reviewed finite-target Coxeter enumeration and the LINS normal-subgroup algorithm. Their completeness and the supplied finite group tables remain computational premises. This note justifies a further restriction when the desired quotient has a prescribed order. It then applies that restriction to the last square diagram in Table 3 of Fernandes–Leemans–Weiss, *An Exploration of Locally Spherical Regular Hypertopes*, fixed 16-page edition, physical page 14.

**Lemma 1.** *Let $N \lhd W$ have finite index $m$. Every subgroup $H$ containing $N$ has index dividing $m$. In particular, every normal overgroup of $N$ is retained by the condition $[W : H] \mid m$.*

*Proof.* The index formula gives $m = [W : H][H : N]$. □

**The search and its dependencies.** The acquired search first supplies the required normal kernels of finite target epimorphisms. Its later operations descend through normal $p$-group sections and take intersections of already known normal subgroups. The finite-target list, module calculations and kernel construction are unchanged. We apply the following restrictions:

1. A finite epimorphic target $Q$ is used only if $|Q| \mid m$.
2. A node $H$ is expanded only if $[W : H] \mid m$.
3. A $p$-section below $H$ is considered only if $p[W : H] \mid m$; a proposed elementary section is retained only if its resulting index $[W : K]$ divides $m$.
4. An intersection is retained only if its computed index divides $m$.

The ordinary bound remains $m$. No early termination is introduced. In particular, "divides $m$" is not replaced by "equals $m$" at an intermediate step.

To justify completeness, fix a desired $N$ and follow the acquired algorithm's completeness argument inside the finite normal-subgroup interval $[N, W]$. Every epimorphic target used in this interval is a quotient of $W/N$, so its order divides $m$. Every intermediate subgroup in a descent contains $N$. Its index and each intervening elementary section therefore satisfy the tests above. If an intersection produces a subgroup containing $N$, both operands also contain $N$, so neither operand nor their intersection is removed. Lemma 1 thus preserves every dependency needed to obtain $N$.

There is also an implementation issue concerning the routine which decides whether a $p$-section is already obtainable by intersections. For any retained $H$, every normal supergroup of $H$ has index dividing $[W : H]$, hence dividing $m$. Its complete supergroup interval is retained. Consequently the minimal-supergroup relations used by that routine have not changed. The same applies to the common-supergroup computation for an intersection. The existing correctness of these optimizations is therefore inherited on the retained interval. It is not inferred from small tests alone.

**Exact implementation and controls.** The program uses LINS's documented option hooks for branch cuts, $p$-quotients, proposed modules and intersections. The initial finite-target routine is the acquired Coxeter routine with one additional order test. The exact presentation is checked there; its involution choices include the identity and proper divisors of bond orders, because a quotient on the way to the desired group may collapse a generator or a bond. All final marked quotients are tested separately for exact orders. The program requires a positive target within the installed table bound, and requires the search bound to equal that same target.

For $A_3, B_3, C_2^3, \widetilde{A}_2, H_3$, the controls compare the restricted search with the full default search. There are 26 target choices and 132 retained normal kernels. Each comparison checks the actual two-way subgroup containment, not merely its index. The $H_3$ cases exercise the nonabelian finite targets. The elementary abelian cases retain many distinct kernels of the same index. A deliberately incorrect restriction to intermediate indices equal to the final target loses kernels, confirming why normal ancestors must be preserved. Invalid target values and a mismatched search bound must be rejected before a search can be credited.

**The square with labels $3, 5, 3, 5$.** Put

$$W = \langle s_1, s_2, s_3, s_4 \mid s_i^2 = 1,\ (s_1s_2)^3 = (s_2s_3)^5 = (s_3s_4)^3 = (s_4s_1)^5 = 1,\ (s_1s_3)^2 = (s_2s_4)^2 = 1\rangle.$$

Every finite regular hypertope with this diagram gives a quotient of this marked presentation. Its four distinguished generators must satisfy

$$H_I \cap H_J = H_{I\cap J}, \qquad H_I = \langle t_i : i \in I\rangle.$$

This necessary condition applies whether a finite-diagram residue is required to be a full spherical Coxeter geometry or a projective quotient.

For target $m = 9720$, the retained normal-kernel indices are

$$1, \quad 2, \quad 60, \quad 120, \quad 4860, \quad 9720,$$

with one kernel at each index. The unique kernel of index 9720 gives the 27-point marked quotient bound in the literal certificate. It has the exact required pair orders and four maximal parabolics of order 120. Nevertheless it fails the intersection property. With indices starting at 1, the literal closure gives

$$|H_{\{1,2,3\}} \cap H_{\{1,2,4\}}| = 12, \qquad |H_{\{1,2\}}| = 6.$$

There is a shorter obstruction in the same tuple. For each omitted $j$, form $z_j = (\prod_{i\neq j} t_i)^5$, with the remaining indices increasing. All four words give the same nonidentity involution $z$. Thus $z$ belongs to every maximal parabolic, whereas repeated application of the intersection property would make their total intersection trivial. The equality and nonidentity of these four words are verified directly on the 27 points; full group closure is unnecessary for this obstruction.

**Theorem 2.** *There is no regular hypertope of square type* $(3, 5, 3, 5)$ *whose type-preserving group has order 9720, conditional on the stated normal quotient enumeration and its retained classification premises.*

*Proof.* Any such hypertope supplies an index-9720 normal kernel of $W$. The complete restricted enumeration contains exactly one. Its marked quotient fails the necessary intersection property as shown above. Relabelling a diagram gives another marking of this same fixed presentation and does not evade its normal-kernel enumeration. □

**Attribution, reuse and scope.** The target order and the displayed group $3^4 \rtimes A_5 \times C_2$ come from the second $(3, 5, 3, 5)$ entry of the source table. The conclusion excludes the order itself and hence does not depend on interpreting an unspecified action in that group name. It is a correction of a published finite claim, not a solution of an open conjecture.

The normal-subgroup algorithm and its finite group tables are due to prior work; the installed documentation attributes the underlying algorithm to Firth's thesis. The finite-target Coxeter traversal, composed module evaluation, complete marked-intersection test and independent literal permutation checker are acquired from earlier reviewed components. The contribution of this note is the explicit justification and use of the divisor restriction, together with its exact source application. Divisibility of finite indices is elementary; no historical priority is claimed for that principle.

The earlier complete search through order 3840 is retained separately. It contains 25 normal kernels, including eleven of index 3840, and gives one valid broad marked hypertope of order 1920 with maximal parabolics of order 60. Its raw finite result and its local interpretation are not silently incorporated into the theorem here. The general bound 9720 previously timed out; that failed computation is preserved and provides no completeness premise. The present theorem concerns the exact target order, not the entire interval below it, and makes no minimum claim.

The certificate checker reruns the bounded restricted computation, compares the ordered marking with the recorded one by an actual marked isomorphism, checks the GAP-to-JSON bridge, and applies the independent integer-permutation closure to all 256 intersections. It separately checks the four displayed central words and malformed input controls. The proof, source table and dependency bytes are retained for distinct review. There is no whole-manuscript credit, new Lean verification, or compression measurement in this component. Other claims and examples in the source remain attached to their own recovery obligations.

# The square $(3, 5, 3, 5)$ through order 3840

We complete the ninth bounded search associated with Table 3 of Fernandes–Leemans–Weiss, *An Exploration of Locally Spherical Regular Hypertopes*, fixed 16-page edition. The normal quotient method, finite target tables and exact marked coset test are acquired from the reviewed components. Their computational premises remain explicit.

Let $W$ have four involutory generators, with consecutive square bond orders $3, 5, 3, 5$ and both diagonal bond orders 2. A regular hypertope with this diagram supplies a normal subgroup of $W$ whose index is its type-preserving group order. This follows from its distinguished chamber involutions and does not assume that its finite-diagram residues are universal spherical geometries.

**Theorem 1.** *Subject to the acquired normal quotient enumeration premises, there is exactly one positive ordered kernel through index 3840. It has index 1920 and four maximal parabolics of order 60. Thus 1920 is the minimum group order when projective finite-diagram residues are allowed. There is no example of order 3840, and no example through that bound with all four universal spherical residues of type $H_3$.*

*Proof.* The complete normal-kernel inventory is

| index | 1 | 2 | 60 | 120 | 660 | 960 | 1320 | 1920 | 3840 |
|---|---|---|---|---|---|---|---|---|---|
| multiplicity | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 7 | 11. |

These are 25 kernels, including the whole group. Exact rank-two subgroups have orders 4, 6 and 10, so a required quotient has order divisible by 60. Only the first two kernels may be omitted by this necessary divisibility test. For each remaining kernel we construct its faithful marked quotient and apply the acquired tests for all 256 subgroup intersections and all 32 base-flag extensions.

Twenty-one of the 23 marked quotients fail the intersection property. This includes all eleven quotients of order 3840. The order-960 quotient passes the intersections but fails the actual incidence test: a specified two-type base flag has ten completions of another type, while its stabilizer orbit has five. Consequently it is not flag-transitive and does not supply the required regular hypertope. The sole passing kernel has order 1920. Its independent literal coset calculation verifies all the flag extensions, so the acquired coset theorem proves that it gives a thin, residually connected, flag-transitive geometry with full type-preserving group of that order.

The four local orders of this positive quotient are all 60. Each local rank-three marked group is a quotient of the order-120 Coxeter group $H_3$, but not the full group. In particular this positive kernel does not satisfy the universal-residue convention. Since every other kernel through 3840 was rejected already under the broader convention, none satisfies the universal-residue condition. □

**The positive witness and its interpretation.** The four involutions act on 12 points. Their complete arrays are retained in the literal certificate; in cycle notation they are

$$t_1 = (1\ 3)(2\ 4)(5\ 6)(7\ 11)(8\ 12)(9\ 10),$$
$$t_2 = (1\ 5)(2\ 6)(3\ 4)(7\ 8)(9\ 11)(10\ 12),$$
$$t_3 = (1\ 2)(3\ 4)(5\ 9)(6\ 10)(7\ 12)(8\ 11),$$
$$t_4 = (1\ 6)(2\ 5)(3\ 7)(4\ 8)(9\ 10)(11\ 12).$$

The literal checker closes the generated group, checks every pair order, forms all sixteen generated subgroups and all their intersections, and constructs every parabolic coset as a set of permutations. Each of the four types has 32 objects. The maximal local quotients have order 60; the earlier spherical classification identifies the corresponding marked geometries as projective $H_3$ residues. No named global group structure is inferred merely from the order 1920.

**Completeness and the retained computation.** The earlier search already returned the 25-kernel inventory and its single positive kernel. The current certificate keeps that execution and its exact dependencies. To make every rejection inspectable, the new output routine also records the compact marked actions for the negative quotients. Its only changes to the acquired caller remove the positive-only output filter and rename the output count. The search, classification inputs, index bound and mathematical tests are unchanged. The checker verifies that exact program difference.

A new bounded execution supplies all 23 actions. An actual isomorphism taking the four generators in order identifies its positive kernel with the earlier one. On subsequent replay the same comparison also checks the whole recorded marked inventory, preserving multiplicities. Every GAP-to-JSON array is read from the current computation. Independent integer permutation closure checks all 23 verdicts, including the order-960 incidence failure. We therefore use neither an abstract group-name match nor an order match as the certificate of a marked quotient.

**Source correction and limits.** The source's first $(3, 5, 3, 5)$ entry has order 3840. The eleven rejected kernels exclude that order under either local convention. With the source's broad finite-diagram meaning, its minimum claim must instead use 1920. If universal spherical residues were intended, the conclusion here is only a lower bound greater than 3840; no existence or least order beyond the bound is asserted.

The second printed order, 9720, is handled in a separate divisor-search component. This note does not repeat or acquire the unsuccessful general search through 9720. The original targets and claims belong to the source authors; the LINS algorithm and classification tables are prior work. The present contribution is the last bounded application, its explicit negative certificates and the correction with its local-scope distinction. It is a finite published-claim correction, not a newly solved open conjecture. Other source obligations, whole-manuscript registration, Lean formalization and compression measurements remain separate.

# Exact finite-target seeds and quotient evaluation for a Coxeter search

Bounded L033 method component

**Abstract**

We replace two operations in the retained LINS normal-subgroup search: finite-target epimorphism enumeration and evaluation of a composed homomorphism. Both changes preserve the complete set of normal kernels. No new hyperbolic quotient, minimum, named group, or completed target bound is asserted by this component.

## 1 The finite-target list

Let

$$W = \langle s_1, \ldots, s_r \mid s_i^2 = 1, (s_i s_j)^{m_{ij}} = 1 \ (i < j) \rangle$$

for a specified finite Coxeter matrix. Our interface checks the literal presentation, including every relator, against this matrix. Fix a finite target group $Q$. Every homomorphism $W \to Q$ corresponds to a tuple $(a_1, \ldots, a_r)$ with $a_i^2 = 1$ and $(a_i a_j)^{m_{ij}} = 1$. Identities and proper divisors of the prescribed orders are allowed at this stage. Surjectivity is exactly $\langle a_1, \ldots, a_r \rangle = Q$.

After choosing a prefix $(a_1, \ldots, a_{k-1})$, its stabilizer under simultaneous inner conjugation is $C_Q(a_1, \ldots, a_{k-1})$. The possible next elements are exactly the square-one elements satisfying the relations to all chosen marks. This set is invariant under that stabilizer. Choose one representative from each of its actual conjugation orbits and continue with the centralizer of the extended prefix. Induction proves that the resulting full tuples represent every inner-conjugacy orbit exactly once. Filtering the full tuples by generation and constructing the actual homomorphism therefore finds every epimorphism up to inner conjugation. Inner-conjugate homomorphisms have the same kernel. Outer-conjugate ones may still have the same kernel; the normal-subgroup graph removes these duplicates. No quotient-isomorphism shortcut is used to remove distinct kernels.

The retained default LINS routine first seeds its graph with all kernels of epimorphisms to the finite targets in its table through the bound. Its installed `findTQuotients.gi` explicitly uses `GQuotients` for this operation. Our initialization inserts precisely the same set of kernels through the existing graph-insertion interface. It then disables only that already completed target-seeding pass. All subsequent prime-power quotient, intersection, and graph operations are unchanged, except the evaluation below. The invocation uses the default options apart from this initialization and target-list filter; no cutting or early termination option is supplied. The caller retains the original index guard. The LINS algorithm and its finite classification tables, through their installed $10^7$ limit, remain explicit computational and classification premises, as in the accepted eight-diagram core. We do not reprove their completeness.

## 2 Evaluation in the actual quotient

At a prime-power step LINS constructs an isomorphism $\iota : H \to \widehat{H}$, where $\widehat{H}$ is finitely presented, and an epimorphism $\mu : \widehat{H} \to E$ onto an elementary abelian group. It obtains the conjugation

matrix by evaluating words $w = xyx^{-1} \in H$ as $\mu(\iota(w))$. Set $\nu = \mu \circ \iota$. Then $\nu(w) = \mu(\iota(w))$ for every such word. Consequently every matrix entry, every invariant submodule, every pullback kernel, and every recursive search remains identical.

The process-local override adds the local variable `Nu`, sets `Nu:=Iso*Mu` in GAP's composition notation, and replaces only that nested image by `Image(Nu,w)`. All other executable statements are literal copies of the pinned installed function. No installed package is edited. GAP's `ghomfp.gi` composition method copies the augmented coset table and maps its generator images through the second map. Rewriting can thus evaluate directly in $E$ instead of building a long intermediate word in $\widehat{H}$. This implementation explains the possible saving; the mathematical equality above does not depend on a timing assumption or an injectivity claim for $\mu$.

**Controls and scope.** A literal traversal of all square-one tuples independently agrees with the inner-orbit traversal for three matrices and six finite groups (18 cases, seven surjective inner orbits). Complete old and new normal kernel inventories agree bijectively as ordered marked quotients for three bounded presentations. During those changed searches all 676 composed conjugation images are also evaluated by the original nested formula and compared exactly. Invalid matrices and mismatched source presentations are rejected. The independent core checker still requires full intersection and actual flag-transitivity tests for positive hypertopes; this method does not weaken them.

The preserved first control driver accidentally captured the replacement function as its reference. Its result is not used for the original-versus-new traversal comparison; the corrected driver reruns all controls while capturing the installed function first. A pair of capped diagnostic profiles shows one index-1920, dimension-71 matrix step taking about 12.7 seconds by nested evaluation and 1.1 seconds by composition. Neither profile completes its target bound, and neither is a new census or general performance theorem. All earlier capped searches remain uncredited.

**Attribution.** The finite targets and surrounding algorithm are the installed LINS 0.9 implementation, based on Firth's normal-subgroup algorithm; the direct composition method is already present in GAP 4.15.1. Stabilizer-orbit traversal is standard. The present author supplies the Coxeter interface, exact substitution proof, code, and bounded controls. The root supplied the original L033 probe; a separate reader supplied the acquired LINS interface audit. No novelty, whole-source, measurement, question, or Lean credit is claimed.

# The two-colour cubic example as an affine Coxeter geometry

Example 2.5 of Fernandes–Leemans–Weiss uses red cubes, yellow cubes, edges and vertices of the ordinary cubic tessellation. We identify its actual incidence structure with the universal coloured-vertex model underlying the acquired halving construction. This argument uses the infinite tessellation directly; it does not set a finite period parameter equal to zero outside the hypotheses of a toroid theorem.

**A coloured-vertex geometry.** In the unit cubic tessellation of $\mathbb{R}^3$, take objects of types 0 and 1 to be the even and odd vertices of $\mathbb{Z}^3$, respectively. Types 2 and 3 are its squares and cubes. Opposite-colour vertices are incident when they differ by a signed coordinate vector. Incidence with a face, or between faces, is containment. Put

$$D = \{v \in \mathbb{Z}^3 : v_1 + v_2 + v_3 \equiv 0 \pmod 2\}, \qquad K = D \rtimes B_3,$$

where $B_3$ is the full signed permutation group.

**Proposition 1.** *This incidence structure is a regular hypertope. Its full type-preserving group is $K$. With the base chamber*

$$0, \quad e_1, \quad [0,1]^2 \times \{0\}, \quad [0,1]^3,$$

*its distinguished involutions are*

$$x = \tau_{e_1+e_2} z_1 z_2 s_1, \qquad s_1, \qquad s_2, \qquad z_3.$$

*Here $s_i$ exchanges coordinates $i, i+1$, $z_i$ reverses coordinate $i$, and composition acts from the rightmost factor first. The diagram has bonds* 02 *and* 12 *labelled 3, and bond* 23 *labelled 4.*

*Proof.* A chamber is an edge, with its two differently coloured endpoints, contained in a square contained in a cube. Every flag extends to such a chain. Indeed its existing squares and cubes are nested; any specified vertices lie in its smallest face. If both vertices occur they form an edge of that face, since they are unit-step neighbours. Extend that edge, or a specified vertex, through the missing face dimensions. With no prescribed face one may choose an incident unit square and cube.

There are exactly two completions of any corank-one flag. For a missing vertex these are the two neighbours of the other vertex in its square. For a missing square they are the two faces of the given cube containing the specified edge. For a missing cube they are the two cubes containing the given square. Thus the geometry is thin.

Its two-type residue incidence graphs are cycles with the following lengths:

| remaining types | 01 | 02 | 12 | 03 | 13 | 23 |
|---|---|---|---|---|---|---|
| cycle length | 4 | 6 | 6 | 4 | 4 | 8. |

For 02 and 12, the residue records the three edges and three faces at a corner of a fixed cube. For 23, it records the four squares and four cubes around a fixed edge. The other cases are the four corners of a square, or the two relevant corners together with the two cubes on that square.

Every three-type residue is connected as well. At a cube the ordinary cube graph connects all its vertex objects, and every square meets it. At a square the four corners and its two containing cubes form a connected graph. At a vertex, its six neighbours are connected through the squares: every two neighbours on different coordinate axes belong to a square through that vertex. Each incident cube meets this connected part. The whole structure is connected because the unit grid connects all vertex objects and every face contains a vertex. This proves residual connectivity, and the two-type calculation gives the diagram.

Translations in $D$ move any even vertex to the origin. Signed coordinate permutations then act simply transitively on the 48 chains from that vertex: six choices of edge, four of containing square and two of cube. Hence $K$ acts regularly on all chambers. The four displayed maps fix exactly the other three objects of the base chamber, and move their own object to its other completion. The acquired universal affine calculation proves that these maps generate $D \rtimes B_3$ with precisely the affine Coxeter presentation stated above. In particular the chamber graph is connected. A type-preserving automorphism fixing a chamber fixes every adjacent chamber by thinness, hence every chamber. Thus there are no additional type-preserving automorphisms. □

**The literal identification with the source.** Colour a source cube $v + [0,1]^3$ red when $\sum v_i$ is even, and yellow otherwise. Use the dual cubic grid with vertices $\frac{1}{2}\mathbf{1} + \mathbb{Z}^3$. A source cube maps to its centre. A source vertex $v$ maps to the dual cube $v + [-\frac{1}{2}, \frac{1}{2}]^3$. A source edge $v + [0,1]e_i$ maps to the dual square whose $i$th coordinate is $v_i + \frac{1}{2}$, and whose other coordinates range from $v_j - \frac{1}{2}$ to $v_j + \frac{1}{2}$.

These maps are bijections onto the respective dual cells. A source vertex is in an edge or cube precisely when the corresponding dual square or vertex is contained in its dual cube. Likewise an edge lies in a source cube precisely when its dual square contains that cube's centre. Finally two source cubes of different colours share a square precisely when their centres are unit-step neighbours. This checks every pair of distinct types, not just chambers.

Translate the dual grid by $-\frac{1}{2}\mathbf{1}$. Source types $(0,2,1,3)$ then become types $(0,1,2,3)$ of the proposition: red cubes become even vertices, yellow cubes odd vertices, edges squares and vertices cubes. Thus the source example is exactly this regular hypertope, with the indicated permutation of types. In the source numbering the bonds 01 and 21 have label 3 and bond 13 has label 4, agreeing with its native diagram.

**Acquired theory, checks and attribution.** The equality and Coxeter presentation

$$\langle x, s_1, s_2, z_3 \rangle = D \rtimes B_3$$

are the universal affine lemma in the accepted generalized-halving calculation. They precede all finite period quotients. The coloured-vertex model is also the universal model underlying that construction. The present addition is the explicit incidence bijection and the proof of the infinite example at the exact interface required by this source.

Small integer-affine checks verify the four displayed transformations, their exact pair orders, their three fixed base objects and their distinct fourth object. The 48 signed coordinate maps give 48 distinct base-vertex chambers. These checks accompany the direct proof; finite sampling is not used to prove residual connectivity of an infinite grid. The example and its red/yellow interpretation belong to Fernandes–Leemans–Weiss. The earlier halving construction and affine calculation retain their stated sources and authorship. This note proves one source example and asserts no new classification, conjecture result, whole-manuscript disposition, formal verification or compression rate.